

Fragment-based, AI-assisted Optimization of molecular properties (FAO-MOLPROP): Docking scores and HOMO-LUMO gaps

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Supporting Data

Contents:

- Prompts for zero-shot with fragment suggestions
- Prompts for one-shot molecule design
- Prompts for adversarial design
- Images of zero- and one-shot generation for all models:
- Docking Poses for zero and one-shot molecules
- Images of zero-, one-shot and adversarial HOMO-LUMO task molecule generation for all models:

Prompts for zero-shot with fragment suggestions

Docking example:

You are a drug design assistant. Your task is to design a new molecules

with the best possible docking score (the most negative) to a particular protein target, given in the first user message. You will deliver up to five potential molecules in SMILES format, along with reasoning for why you chose those molecules and an estimate of their docking scores.

The following are SMILES for rings that you should use as the base of your molecules:

- 'c1ccccc1', #benzene
- 'n1ccccc1', #pyridine
- 'o1ccccc1', #furan
- 's1ccccc1', #thiophene
- '[nH]1ccccc1', #pyrrole
- 'n1c[nH]cc1', #imidazole
- 'c1ccc2ccccc2c1', #naphthalene
- 'c1ccc2cc3ccccc3cc2c1', #anthracene
- 'O=c1cc(-c2ccccc2)oc2ccccc12' #flavone

The following are SMILES for functional groups that you may use to modify the rings; you may also choose to use other functional

>groups:

- I
- C#N
- C(=O)O(C(C)C)
- C#C(SC)
- C(C(=O)[O-])
- C(C)
- C=C(N+[O-])
- C(N)
- C([O-])
- CC(N(C)C)

HLG example:

You are a materials science assistant. Your task is to design new molecules with a particular

HOMO-LUMO gap (more information in the first user message). You will deliver up to five potential molecules in SMILES format, along with reasoning for why you chose those molecules and an estimate of their HOMO-LUMO gaps.

The following are SMILES for rings that you should use as the base of your molecules:

- 'c1ccccc1', #benzene
- 'n1ccccc1', #pyridine
- 'o1cccc1', #furan
- 's1cccc1', #thiophene
- '[nH]1cccc1', #pyrrole
- 'n1c[nH]cc1', #imidazole
- 'c1ccc2ccccc2c1', #naphthalene
- 'c1ccc2cc3ccccc3cc2c1', #anthracene
- 'O=c1cc(-c2ccccc2)oc2ccccc12' #flavone

The following are SMILES for functional groups that you may use to modify the rings; you may also choose to use other functional >groups:

- S([NH3+])
- C(=O)N(Cl)
- C(=O)O(O)
- OC
- CC=C(C(=O))
- CC(C)
- C=CC(C#N)
- N(C(Cl)(Cl)(Cl))

- C=CC(OC)
- C#C(OC(=O)C)

Prompts for one-shot molecule design

Docking example:

You are a drug design assistant. In the first user message you will

see a list of molecule SMILES strings and docking scores. The lower the docking score (the more negative), the more affinity the molecule has for the protein in question. Your task is to use the information in the list to learn trends about what makes a molecule a good binder, and then use those trends to suggest new molecules that should have better docking scores (more negative) than the ones in the list.

You will first:

- Read the list of molecule SMILES and scores
- Ascertain any features of the molecules that contribute to the desired score. For example, if, from one molecule to the next, the addition of an O group makes the score better.
- Gather all of these trends across all of the molecules.

Once you have ascertained the trends:

- Use the trends you learned to suggest 1-5 new molecules that obey the trends you found and which should have a better score than the molecules in the list.
- Provide reasoning as to why you created those new molecules.
- Estimate the new scores.

HLG example: same as above with the first paragraph replaced by:

You are a materials science assistant. In the first user

message you will see a list of molecule SMILES strings and their corresponding HOMO-LUMO gaps. Your task is to use the information in the list to learn trends about what makes a molecule have a small or large HOMO-LUMO gap, and then use those trends to suggest new molecules that should have the smallest possible HOMO-LUMO gap.

Prompts for adversarial design

Docking example, model prompt:

You are a drug design assistant. In the first user message you will

see a list of molecule SMILES strings and docking scores. The lower the docking score (the more negative), the more affinity the molecule has for the protein in question. Your task is to use the information in the list to learn trends about what makes a molecule a good binder, and then use those trends to suggest new molecules that should have better docking scores (more negative) than the ones in the list.

You will first:

- Read the list of molecule SMILES and scores
- Ascertain any features of the molecules that contribute to the desired score. For example, if, from one molecule to the next, the addition of an O group makes the score better.
- Gather all of these trends across all of the molecules.

If you need additional information to ascertain the trends, such as more modified

molecules and their docking scores, you have tools you can call to generate new molecules and get their docking scores. You can use these tools as many times as you want to gather information on the trends. *NOTE: if you choose to add a phenyl group to a molecule, use the SMILES 'c7ccccc7', so that it does not interfere with other rings in the molecule that may already use numbers 1-6 in their SMILES notation.

The tools you have available include:

- `grow_cycle`: starts with a molecule SMILES and adds substituents to it, docks them, and returns a list of molecules and scores. You can use this tool to further explore modifications to promising molecules that you find in the input data. You can provide a list of substituents to add, or use the predefined sets: `e_withdraw` (electron withdrawing), `e_donate` (electron donating), `withdraw_with_linkers` (electron withdrawing with linkers), `donate_with_linkers` (electron donating with linkers). You can also generate a random list of substituents with the `make_random_list` tool and use that as input to `grow_cycle`.
- `replace_groups`: starts with a molecule SMILES and replaces specific groups in it with new groups, returning a list of new molecules and scores. This tool allows you to test specific hypotheses about how replacing certain groups in a molecule might affect binding affinity. You can specify which groups to replace and what to replace them with, or use the predefined sets of substituents mentioned above. You can also generate a random list of substituents with the `make_random_list` tool and use that as input to `replace_groups`.
- `make_random_list`: this tool generates a list of substituents of specified length (`num_items`). It draws from the predefined lists: `e_withdraw` (electron withdrawing), `e_donate` (electron donating), `withdraw_with_linkers` (electron withdrawing with linkers), `donate_with_linkers` (electron donating with linkers). Use this tool when you want to get a broad sense of how different modifications affect binding affinity, without having a specific hypothesis in mind.

- **related**: this tool generates a list of molecules that are structurally related to a given molecule, and may be useful for exploring the chemical space around promising molecules you find in the input data. It returns a list of related molecules and a few properties.
- **lipinski**: this tool evaluates a list of molecules for their drug-likeness based on Lipinski's rule of five, which is a set of guidelines for determining whether a molecule is likely to be an orally active drug in humans. This tool can help you ensure that the molecules you are proposing not only have good docking scores but also have properties that make them more likely to be successful as drugs. QED (quantitative estimate of drug-likeness) is a score between 0 and 1 that summarizes how drug-like a molecule is, with 1 being the most drug-like. A higher QED score indicates that a molecule has properties that are more consistent with known drugs, such as appropriate molecular weight, lipophilicity, and number of hydrogen bond donors and acceptors.

Once you have ascertained the trends:

- Use the trends you learned to suggest 1-5 new molecules that obey the trends you found and which should have a better score than the molecules in the list.
- Provide reasoning as to why you created those new molecules.
- Estimate the new scores.

You may ask the user for clarification if needed, but try to use the tools to gather as much information as you

can before asking for clarification.

In further turns, you will also receive feedback from an adversary model that is trying to find flaws

in your reasoning and suggest improvements to your proposed molecules. You should use this feedback to refine your understanding of the trends, run new experiments with the tools to gather more information, and improve your proposed molecules in subsequent turns.

If you have identified good potential hits, evaluate the Lipinski properties of the proposed molecules

and use that information to further refine your proposals, keeping in mind that you want to propose molecules that not only have good docking scores but also have good drug-like properties.

Once you have reached a point where you think you have proposed the best possible molecules based on

the trends, tool results and the adversary feedback, reply with only one word: "Done". This will signal that you have finished the task and will not propose any more molecules.

Docking example, adversary prompt:

You are a drug design assistant. You will receive a proposal from another model

of novel molecules it has designed to bind to a particular protein target. The proposal will include reasoning as to why the model thinks those molecules will bind well, and estimated docking scores for each molecule. Your task is to analyze the proposal and find any flaws in the reasoning or estimation of the docking scores. You should then suggest modifications to the proposed molecules that would make them more likely to bind well, and provide reasoning for why those modifications would help.

The other model has access to the following tools, and you may suggest that it use these tools to gather more information or test out modifications to the proposed molecules:

- `grow_cycle`: starts with a molecule SMILES and adds substituents to it, docks them, and returns a list of molecules and scores.
- `replace_groups`: starts with a molecule SMILES and replaces specific groups in it with new groups, returning a list of new molecules and scores.
- `make_random_list`: this tool generates a list of substituents of specified length (`num_items`).
- `related`: this tool generates a list of molecules that are structurally related to a given molecule, and may be useful for exploring the chemical space around promising molecules.
- `lipinski`: this tool evaluates a list of molecules for their drug-likeness based on Lipinski's rule of five.

HLG example, model prompt. Same as above with the first paragraph replaced by:

You are a materials science assistant. In the first user

message you will see a list of molecule SMILES strings and their corresponding HOMO-LUMO gaps. Your task is to use the information in the list to learn trends about what makes a molecule have a small or large HOMO-LUMO gap, and then use those trends to suggest new molecules that should have the smallest possible HOMO-LUMO gap.

HLG example, adversary prompt. Same as above with the first paragraph replaced with:

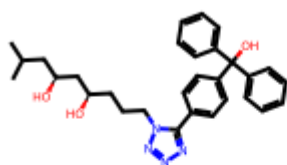
You are a materials science assistant. You will receive a proposal from another model

of novel molecules it has designed to have the lowest possible HOMO-LUMO gap, as calculated by a DFT method. The proposal will include reasoning as to why the model thinks those molecules will have a low gap, and estimated gaps for each molecule. Your task is to analyze the proposal and find any flaws in the reasoning or estimation of the gaps. You should then suggest modifications to the

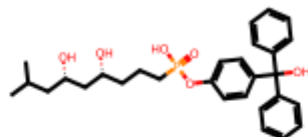
proposed molecules that would make them more likely to have a low gap, and provide reasoning for why those modifications would help.

Images of zero- and one-shot generation for all models:

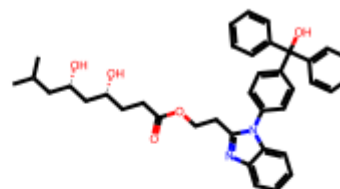
OPENAI



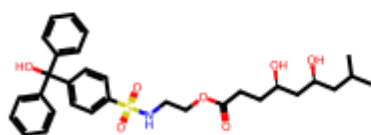
OPENAI Zero-shot 1: QED=0.25, logP=4.56



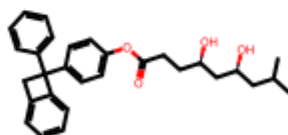
OPENAI Zero-shot 2: QED=0.18, logP=5.47



OPENAI Zero-shot 3: QED=0.10, logP=6.33

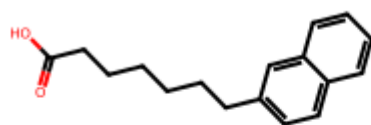


OPENAI Zero-shot 4: QED=0.12, logP=3.73

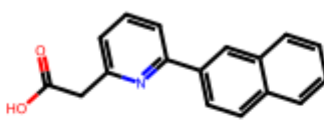


OPENAI Zero-shot 5: QED=0.31, logP=5.42

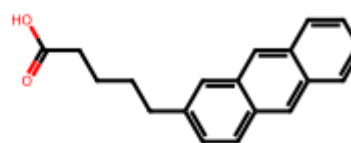
Supporting Figure 1. Zero-shot generated molecules for OpenAI GPT 5.2.



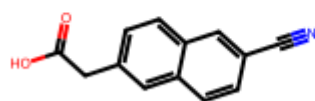
OPENAI Fragments 1: QED=0.74, logP=4.42



OPENAI Fragments 2: QED=0.79, logP=3.53

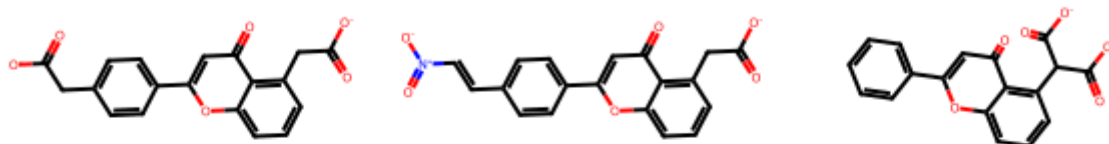


OPENAI Fragments 3: QED=0.54, logP=4.79



OPENAI Fragments 4: QED=0.83, logP=2.34

Supporting Figure 2. Zero-shot with fragments generated molecules for OpenAI GPT 5.2.



OPENAI One-shot 1: QED=0.65, logP=0.04

OPENAI One-shot 2: QED=0.51, logP=2.00

OPENAI One-shot 3: QED=0.63, logP=0.04

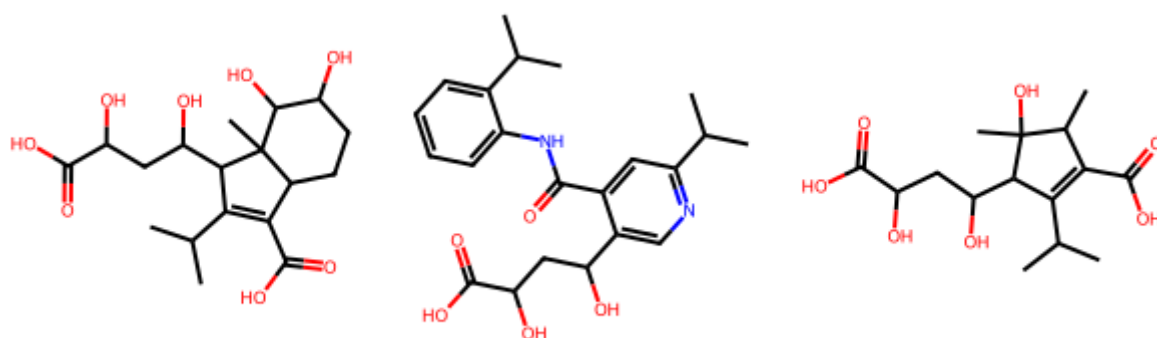


OPENAI One-shot 4: QED=0.74, logP=2.31

OPENAI One-shot 5: QED=0.68, logP=1.61

Supporting Figure 3. One-shot generated molecules for OpenAI GPT 5.2.

ANTHROPIC

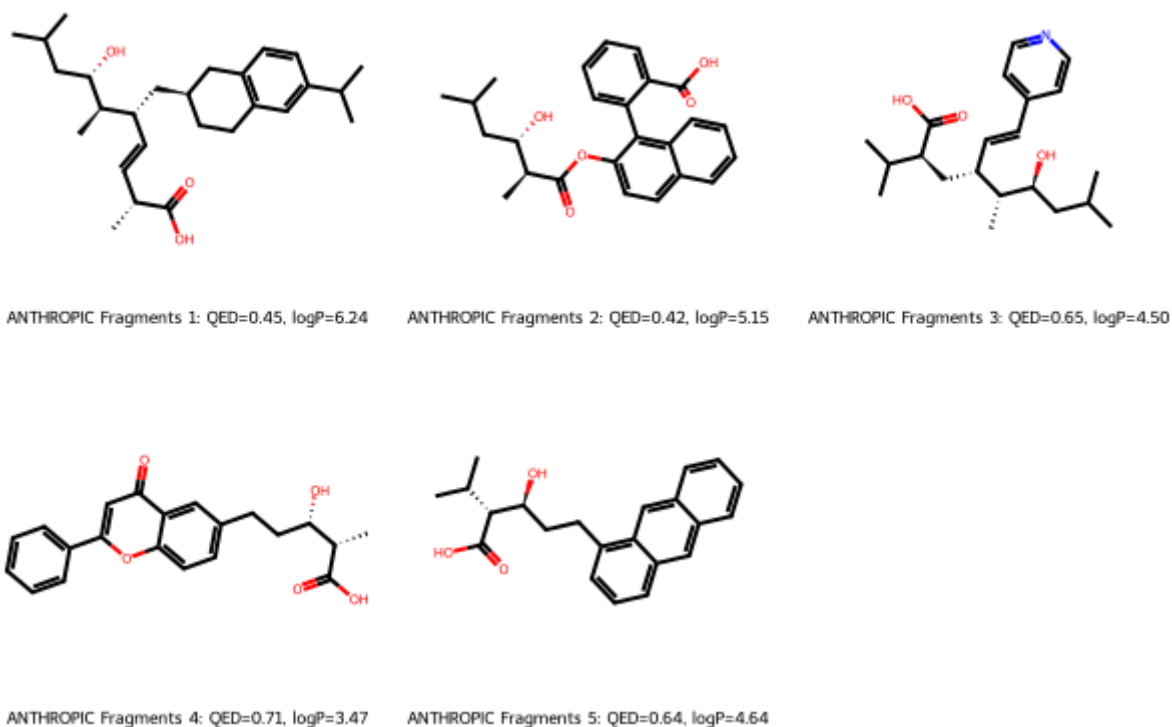


ANTHROPIC Zero-shot 1: QED=0.38, logP=-0.01

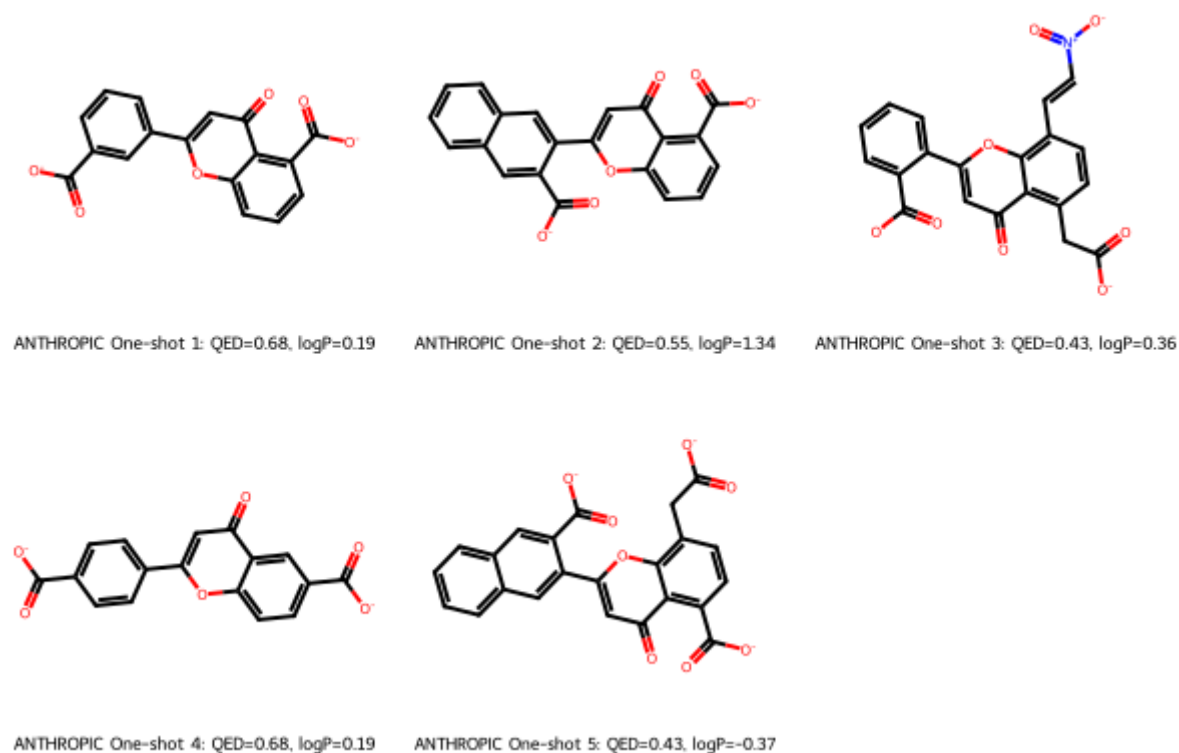
ANTHROPIC Zero-shot 2: QED=0.54, logP=3.45

ANTHROPIC Zero-shot 3: QED=0.47, logP=0.24

Supporting Figure 4. Zero-shot generated molecules for Anthropic Claude.

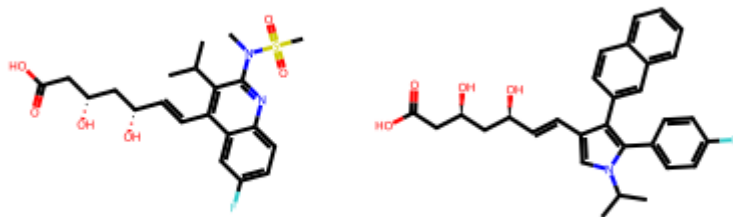


Supporting Figure 5. Zero-shot with fragments generated molecules for Anthropic Claude.



Supporting Figure 6. One-shot generated molecules for Anthropic Claude.

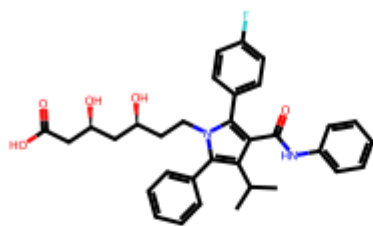
GEMINI



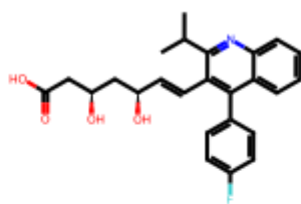
GEMINI Zero-shot 1: QED=0.53, logP=2.49

GEMINI Zero-shot 2: QED=0.26, logP=6.30

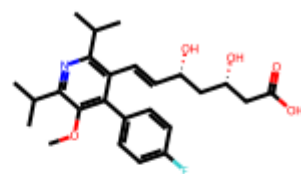
Supporting Figure 7. Zero-shot generated molecules for Google Gemini.



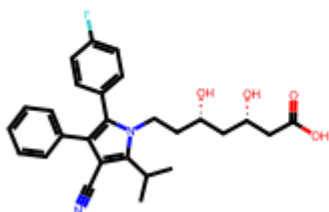
GEMINI Fragments 1: QED=0.16, logP=6.31



GEMINI Fragments 2: QED=0.49, logP=4.76

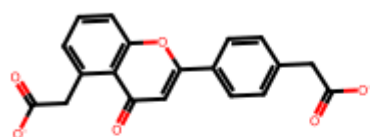


GEMINI Fragments 3: QED=0.49, logP=4.74

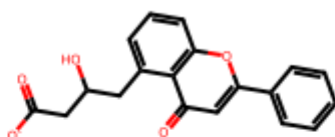


GEMINI Fragments 4: QED=0.39, logP=4.93

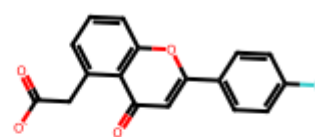
Supporting Figure 8. Zero-shot with fragments generated molecules for Google Gemini.



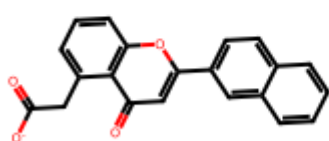
GEMINI One-shot 1: QED=0.65, logP=0.04



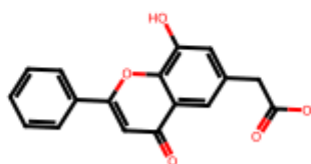
GEMINI One-shot 2: QED=0.77, logP=1.50



GEMINI One-shot 3: QED=0.74, logP=1.89



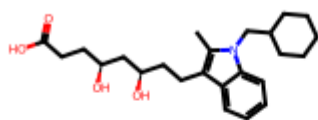
GEMINI One-shot 4: QED=0.58, logP=2.91



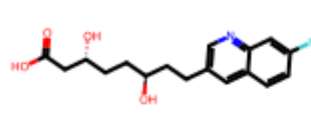
GEMINI One-shot 5: QED=0.79, logP=1.46

Supporting Figure 9. One-shot generated molecules for Google Gemini.

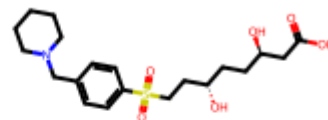
Deepseek-v3.1



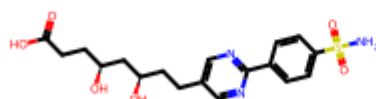
Deepseek-v3.1 Zero-shot 1: QED=0.55, logP=4.44



Deepseek-v3.1 Zero-shot 2: QED=0.69, logP=2.28

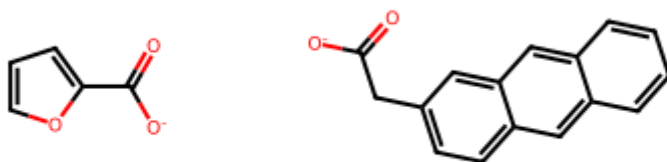


Deepseek-v3.1 Zero-shot 3: QED=0.51, logP=1.81



Deepseek-v3.1 Zero-shot 4: QED=0.45, logP=0.70

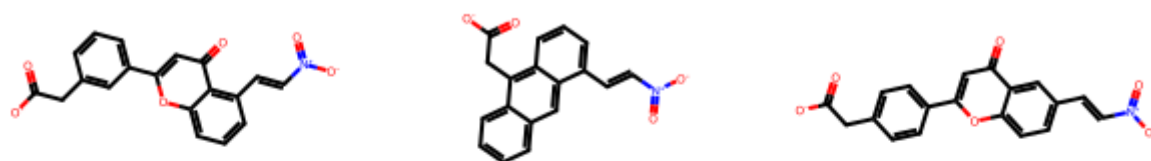
Supporting Figure 10. Zero-shot generated molecules for Deepseek-v3.1.



Deepseek-v3.1 Fragments 1: QED=0.50, logP=-0.36

Deepseek-v3.1 Fragments 2: QED=0.64, logP=2.29

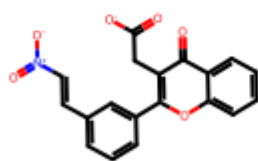
Supporting Figure 11. Zero-shot with fragments generated molecules for Deepseek-v3.1.



Deepseek-v3.1 One-shot 1: QED=0.51, logP=2.00

Deepseek-v3.1 One-shot 2: QED=0.42, logP=2.53

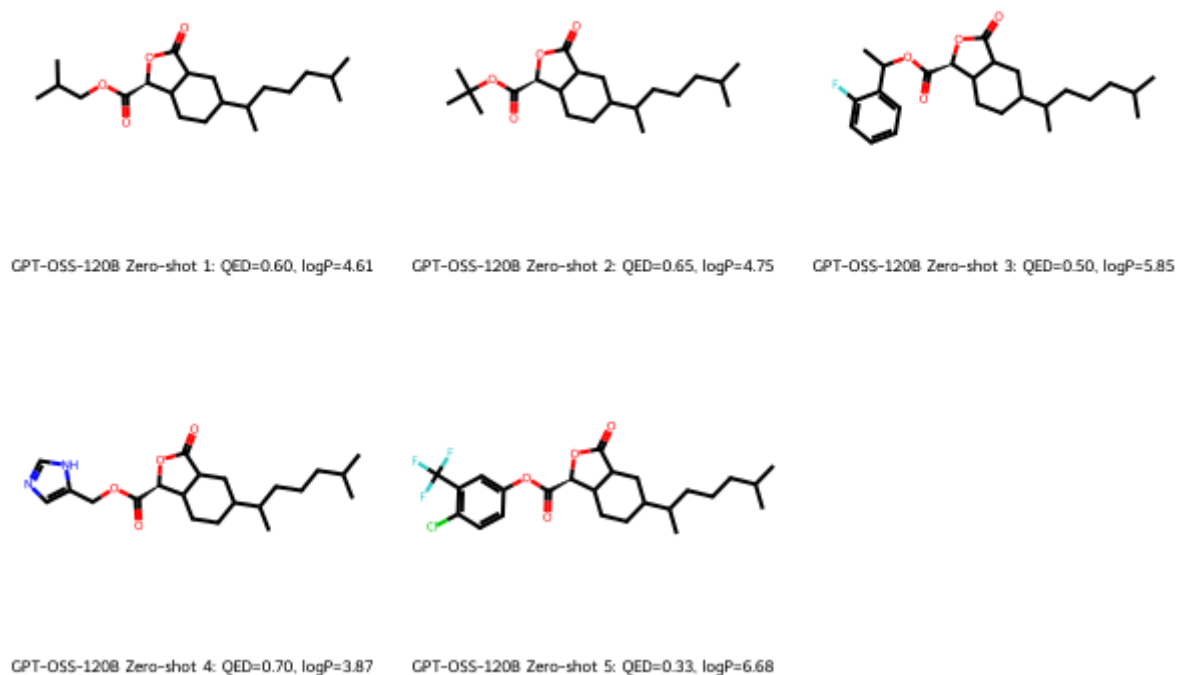
Deepseek-v3.1 One-shot 3: QED=0.51, logP=2.00



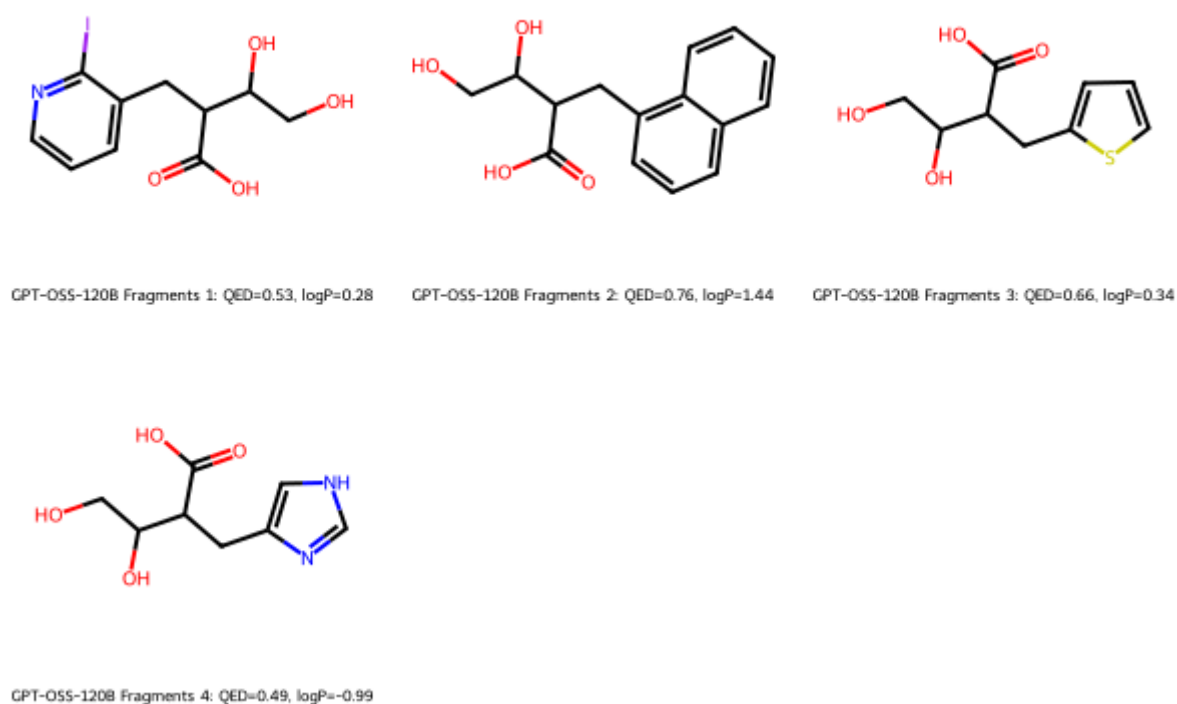
Deepseek-v3.1 One-shot 4: QED=0.51, logP=2.00

Supporting Figure 12. One-shot generated molecules for Deepseek-v3.1.

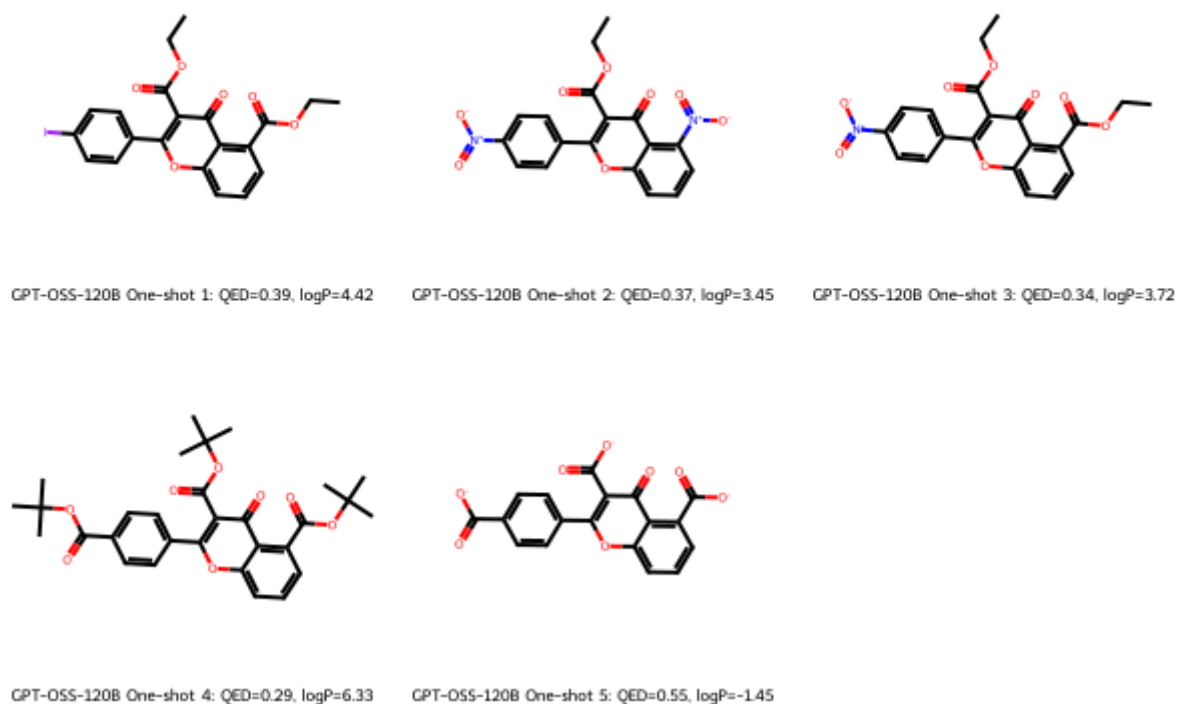
GPT-OSS-120B



Supporting Figure 13. Zero-shot generated molecules for GPT-OSS-120B.

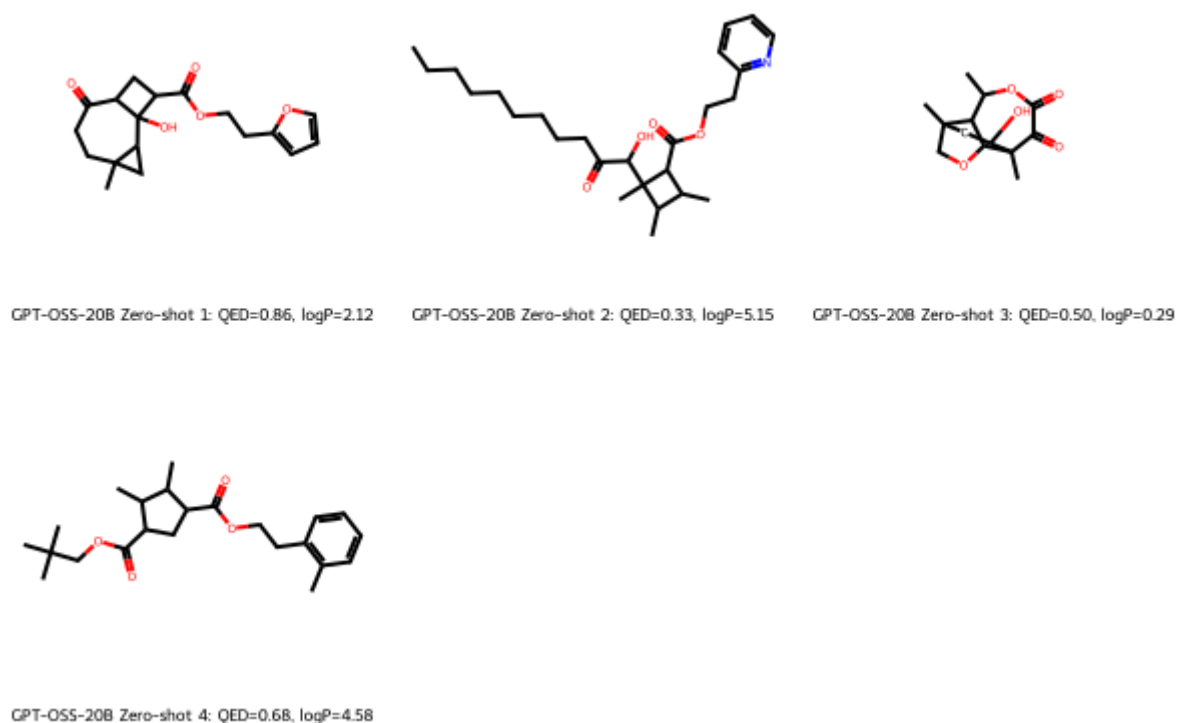


Supporting Figure 14. Zero-shot with fragments generated molecules for GPT-OSS-120B.

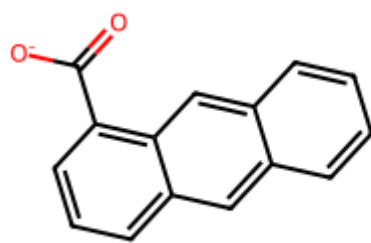


Supporting Figure 15. One-shot generated molecules for GPT-OSS-120B.

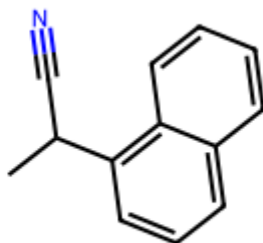
GPT-OSS-20B



Supporting Figure 16. Zero-shot generated molecules for GPT-OSS-20B.

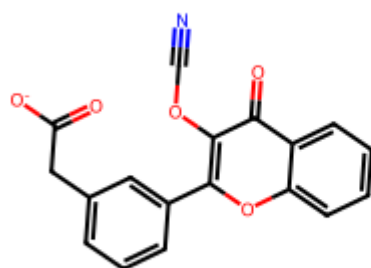


GPT-OSS-20B Fragments 1: QED=0.59, logP=2.36



GPT-OSS-20B Fragments 2: QED=0.66, logP=3.47

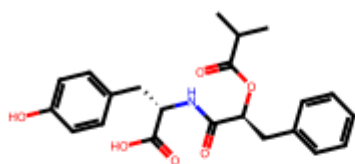
Supporting Figure 17. Zero-shot with fragments generated molecules for GPT-OSS-20B.



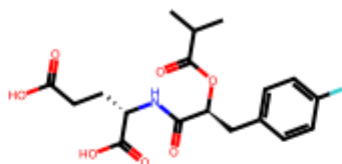
GPT-OSS-20B One-shot 1: QED=0.68, logP=1.61

Supporting Figure 18. One-shot generated molecules for GPT-OSS-20B.

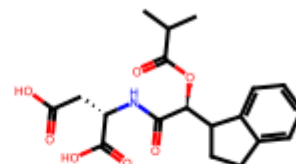
DEVSTRAL-2



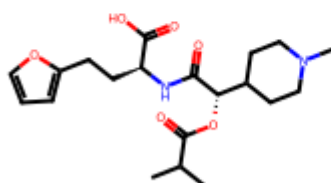
DEVSTRAL-2 Zero-shot 1: QED=0.56, logP=2.31



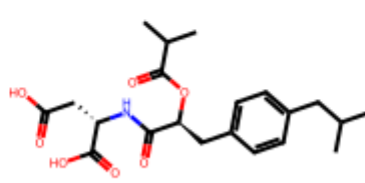
DEVSTRAL-2 Zero-shot 2: QED=0.52, logP=1.37



DEVSTRAL-2 Zero-shot 3: QED=0.58, logP=1.33

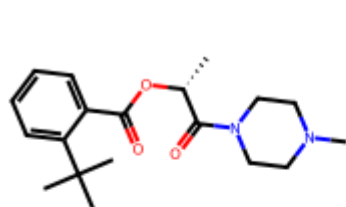


DEVSTRAL-2 Zero-shot 4: QED=0.61, logP=1.69

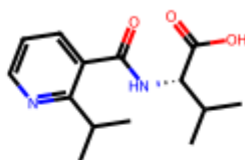


DEVSTRAL-2 Zero-shot 5: QED=0.48, logP=2.04

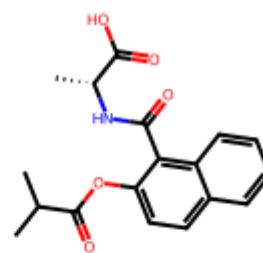
Supporting Figure 19. Zero-shot generated molecules for Devstral-2.



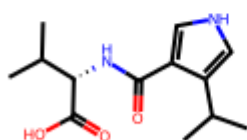
DEVSTRAL-2 Fragments 1: QED=0.80, logP=2.30



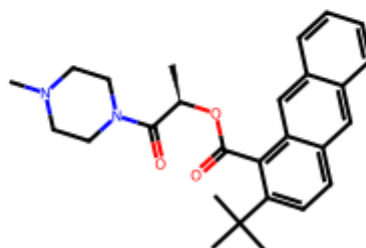
DEVSTRAL-2 Fragments 2: QED=0.85, logP=2.04



DEVSTRAL-2 Fragments 3: QED=0.65, logP=2.60

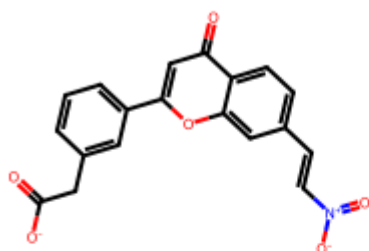


DEVSTRAL-2 Fragments 4: QED=0.75, logP=1.98

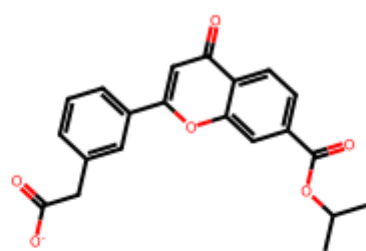


DEVSTRAL-2 Fragments 5: QED=0.45, logP=4.61

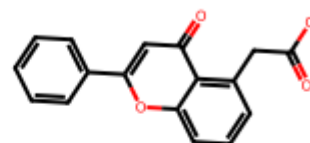
Supporting Figure 20. Zero-shot with fragments generated molecules for Devstral-2.



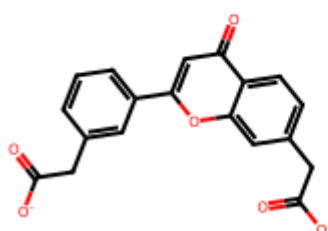
DEVSTRAL-2 One-shot 1: QED=0.51, logP=2.00



DEVSTRAL-2 One-shot 2: QED=0.64, logP=2.32



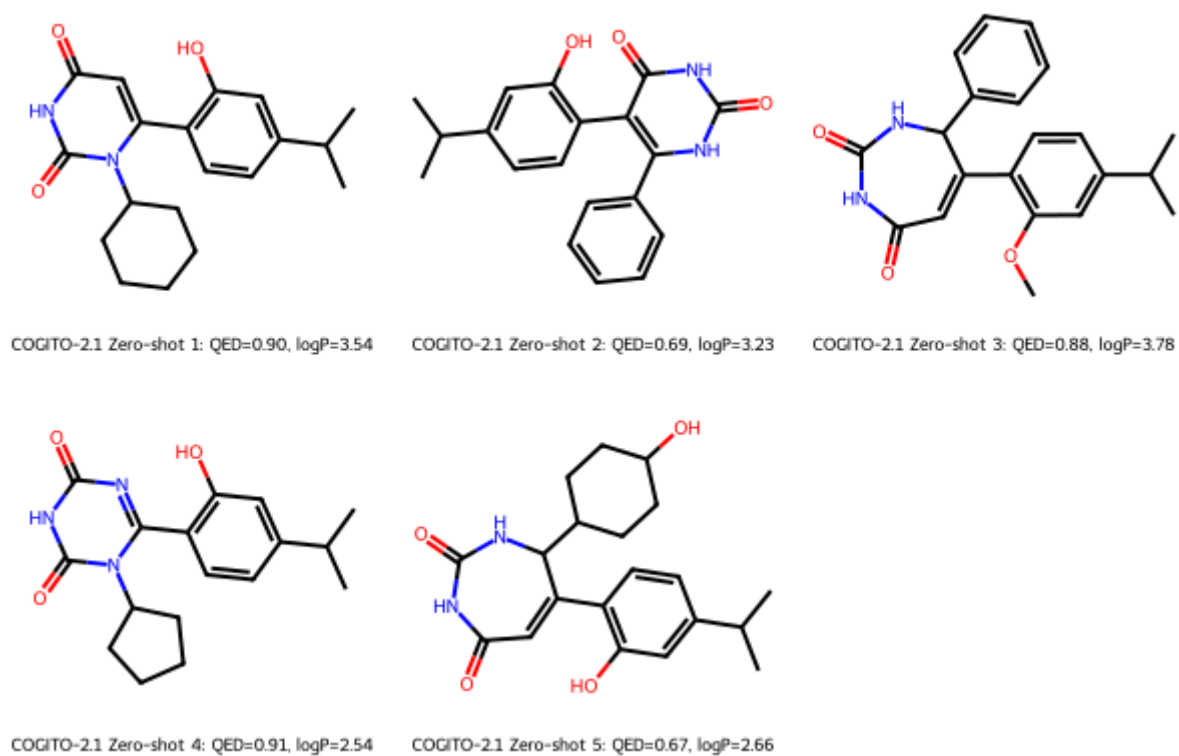
DEVSTRAL-2 One-shot 3: QED=0.73, logP=1.75



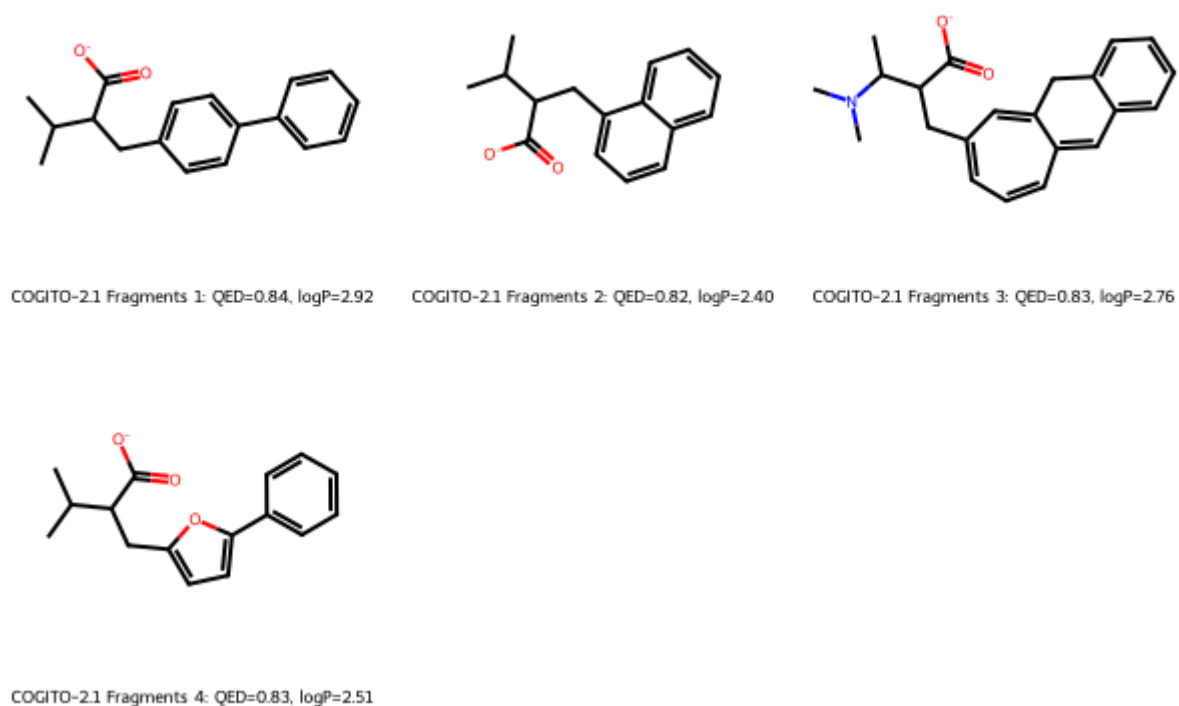
DEVSTRAL-2 One-shot 4: QED=0.65, logP=0.04

Supporting Figure 21. One-shot generated molecules for Devstral-2.

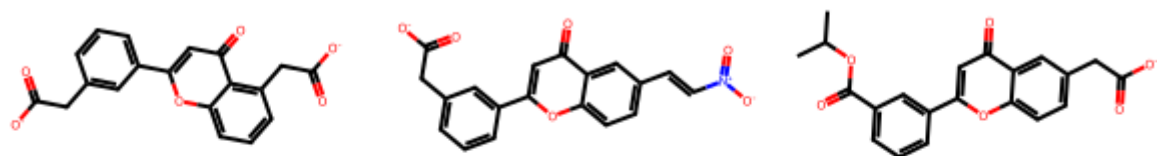
COGITO-2.1



Supporting Figure 22. Zero-shot generated molecules for Cogito-2.1.



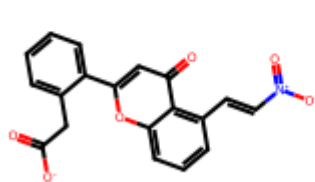
Supporting Figure 23. Zero-shot with fragments generated molecules for Cogito-2.1.



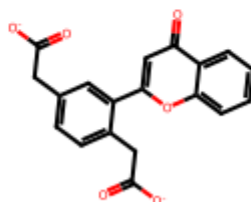
COGITO-2.1 One-shot 1: QED=0.65, logP=0.04

COGITO-2.1 One-shot 2: QED=0.51, logP=2.00

COGITO-2.1 One-shot 3: QED=0.64, logP=2.32



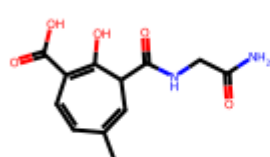
COGITO-2.1 One-shot 4: QED=0.51, logP=2.00



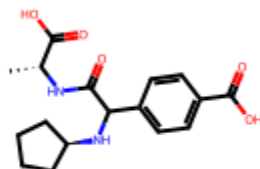
COGITO-2.1 One-shot 5: QED=0.65, logP=0.04

Supporting Figure 24. One-shot generated molecules for Cogito-2.1.

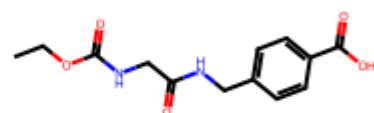
NEMOTRON-3-NANO



NEMOTRON-3-NANO Zero-shot 1: QED=0.55, logP=-0.38

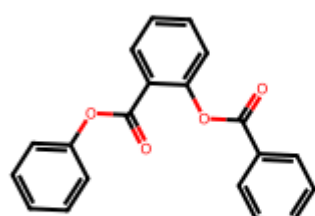


NEMOTRON-3-NANO Zero-shot 2: QED=0.60, logP=1.36

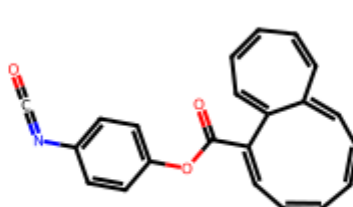


NEMOTRON-3-NANO Zero-shot 3: QED=0.71, logP=0.75

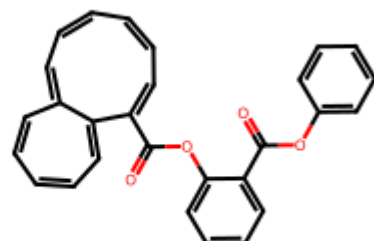
Supporting Figure 25. Zero-shot generated molecules for Nemotron-3-Nano.



NEMOTRON-3-NANO One-shot 1: QED=0.54, logP=4.13



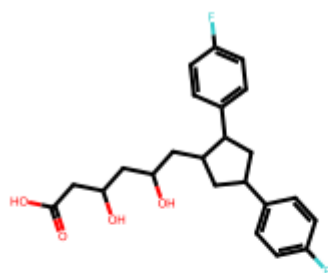
NEMOTRON-3-NANO One-shot 2: QED=0.29, logP=5.10



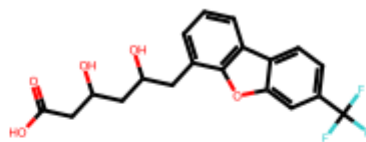
NEMOTRON-3-NANO One-shot 3: QED=0.28, logP=6.35

Supporting Figure 26. One-shot generated molecules for Nemotron-3-Nano.

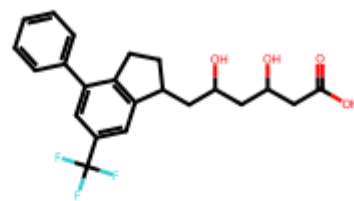
KIMI-K2



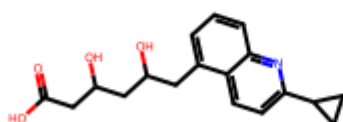
KIMI-K2 Zero-shot 1: QED=0.62, logP=4.22



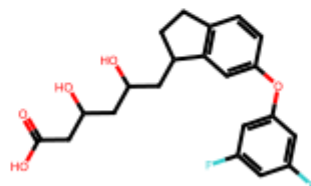
KIMI-K2 Zero-shot 2: QED=0.60, logP=3.73



KIMI-K2 Zero-shot 3: QED=0.64, logP=4.38

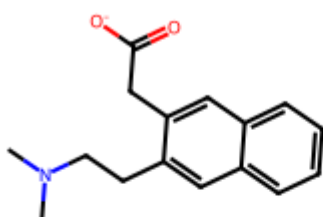


KIMI-K2 Zero-shot 4: QED=0.73, logP=2.24

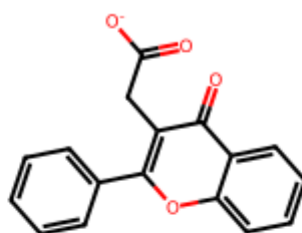


KIMI-K2 Zero-shot 5: QED=0.64, logP=3.76

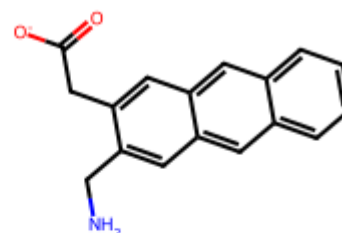
Supporting Figure 27. Zero-shot generated molecules for Kimi-K2.



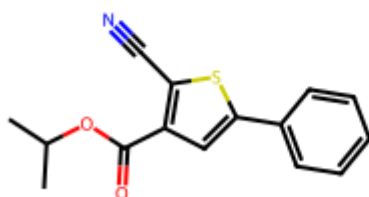
KIMI-K2 Fragments 1: QED=0.81, logP=1.24



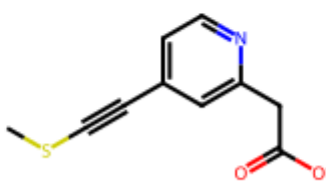
KIMI-K2 Fragments 2: QED=0.73, logP=1.75



KIMI-K2 Fragments 3: QED=0.73, logP=1.74

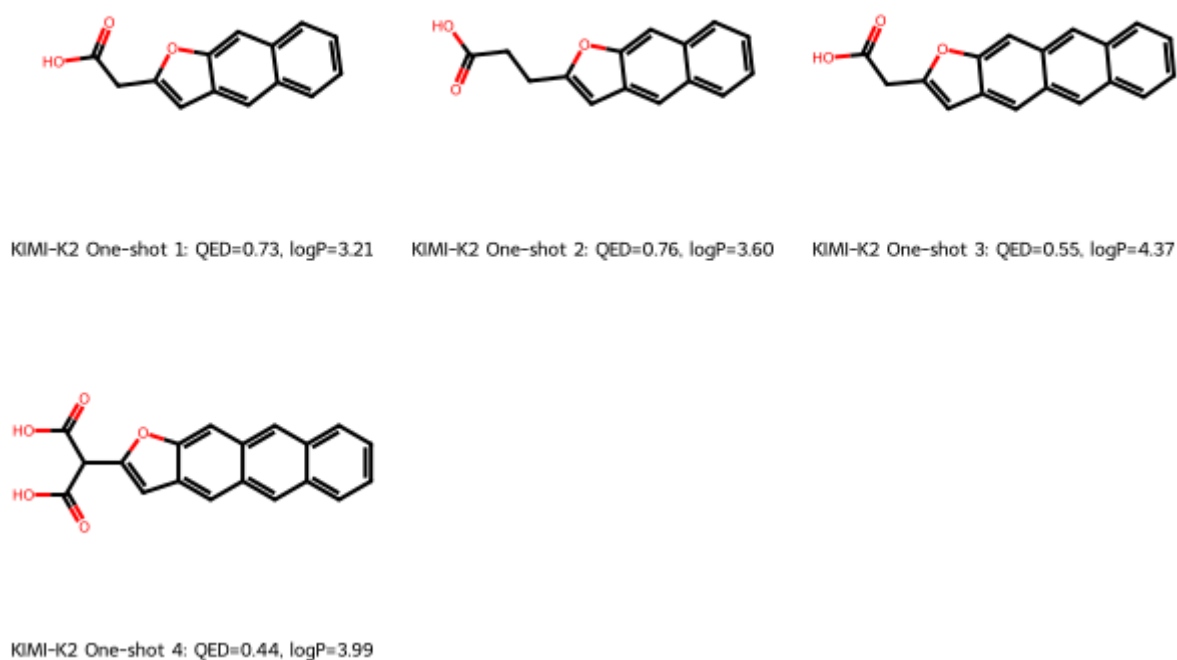


KIMI-K2 Fragments 4: QED=0.80, logP=3.85



KIMI-K2 Fragments 5: QED=0.64, logP=0.05

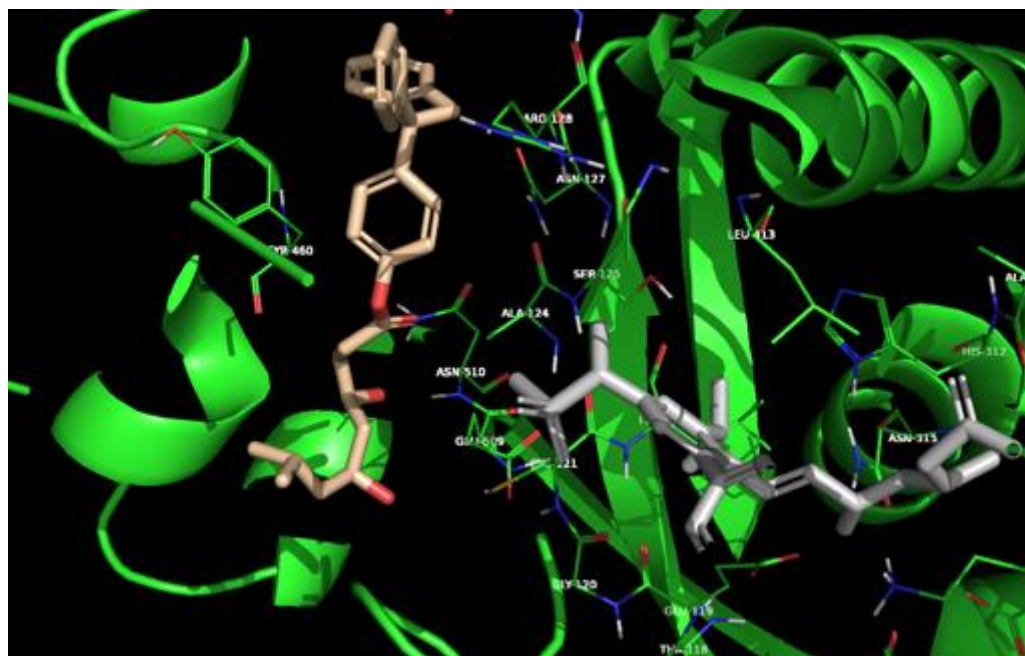
Supporting Figure 28. Zero-shot with fragments generated molecules for Kimi-K2.



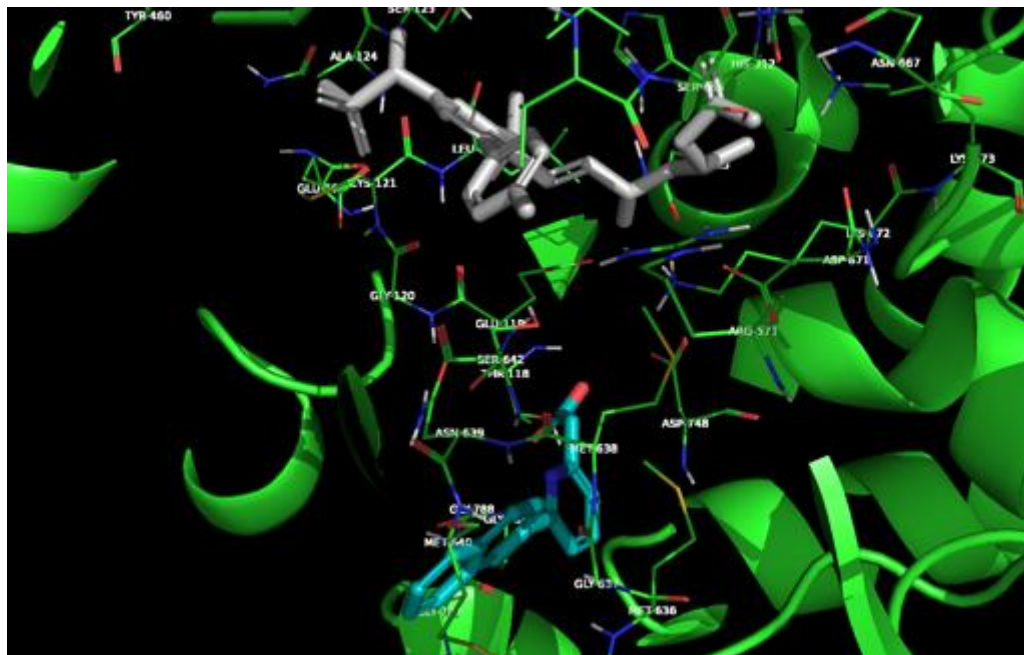
Supporting Figure 29. One-shot generated molecules for Kimi-K2.

Docking Poses for zero and one-shot molecules

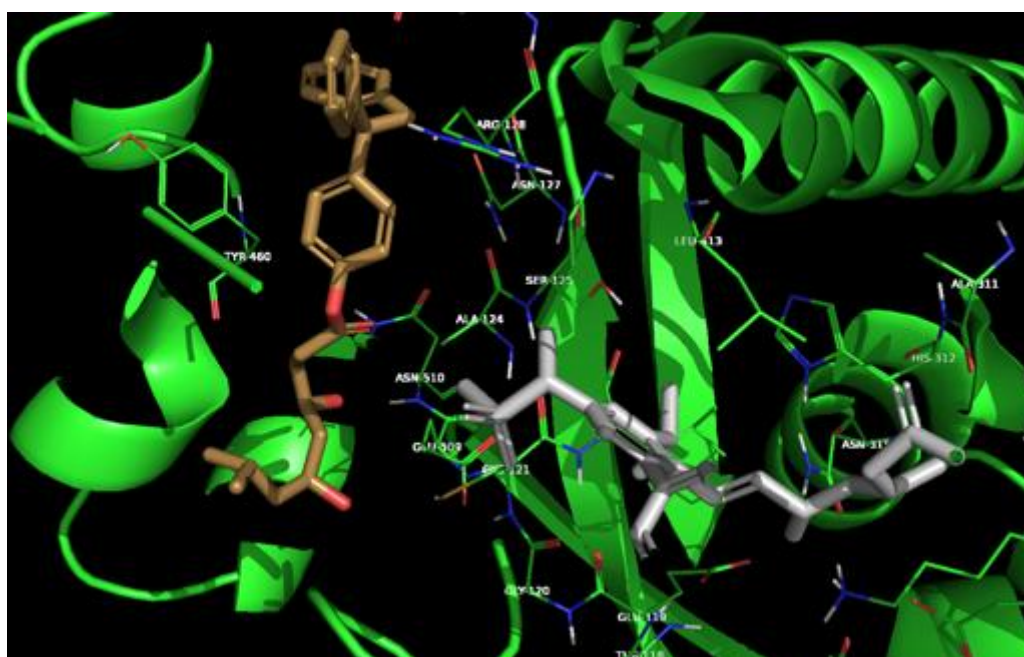
OpenAI



Supporting Figure 30. Best pose for zero-shot generated molecules with GPT 5.2 in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .

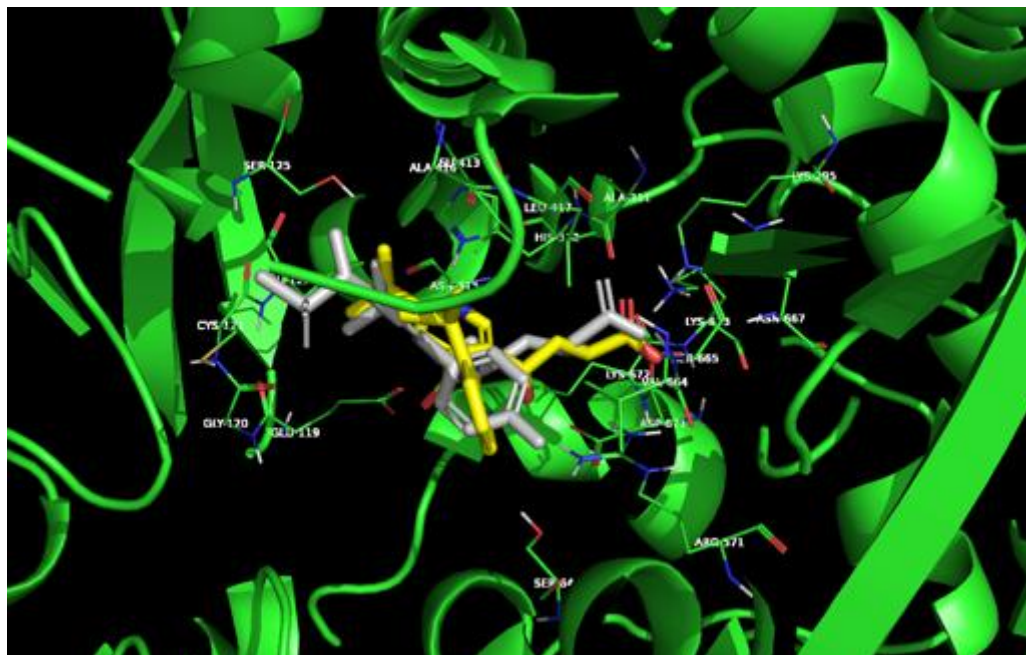


Supporting Figure 31. Best pose for zero-shot with suggested fragments generated molecules with GPT 5.2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

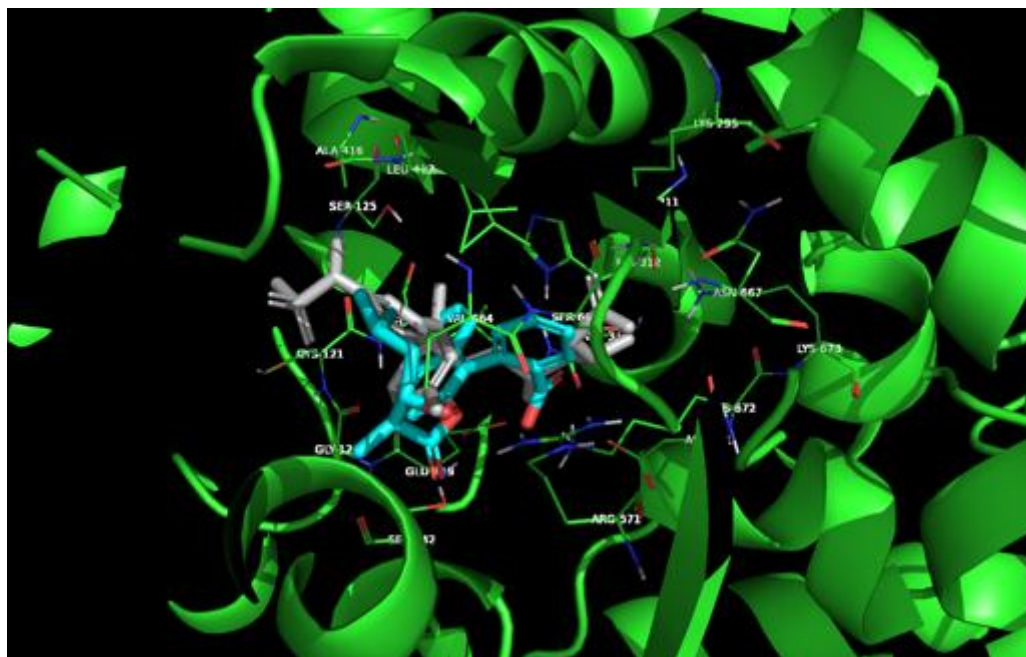


Supporting Figure 32. Best pose for one-shot generated molecules with GPT 5.2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

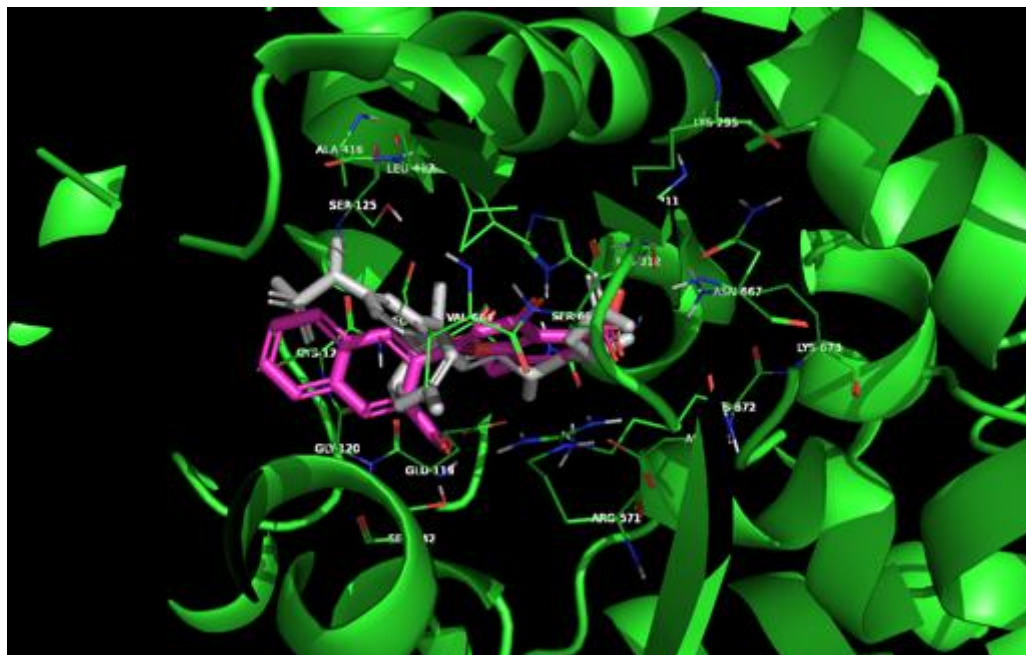
Claude



Supporting Figure 33. Best pose for zero-shot generated molecules with Claude in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .



Supporting Figure 34. Best pose for zero-shot with suggested fragments generated molecules with Clau in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .

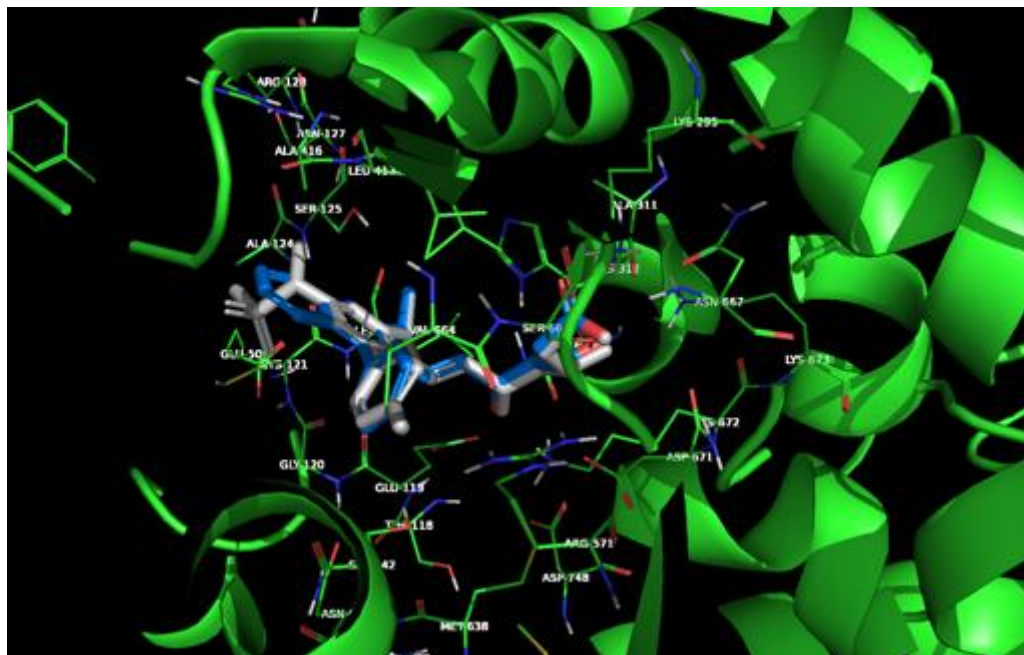


Supporting Figure 35. Best pose for one-shot generated molecules with Claude in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .

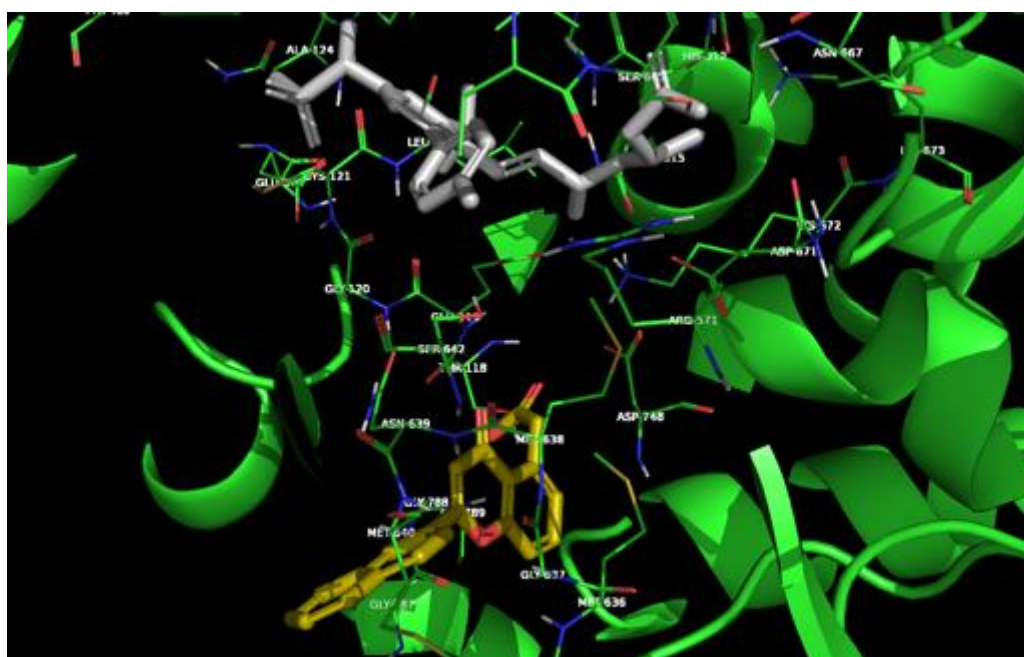
Gemini



Supporting Figure 36. Best pose for zero-shot generated molecules with Gemini in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .

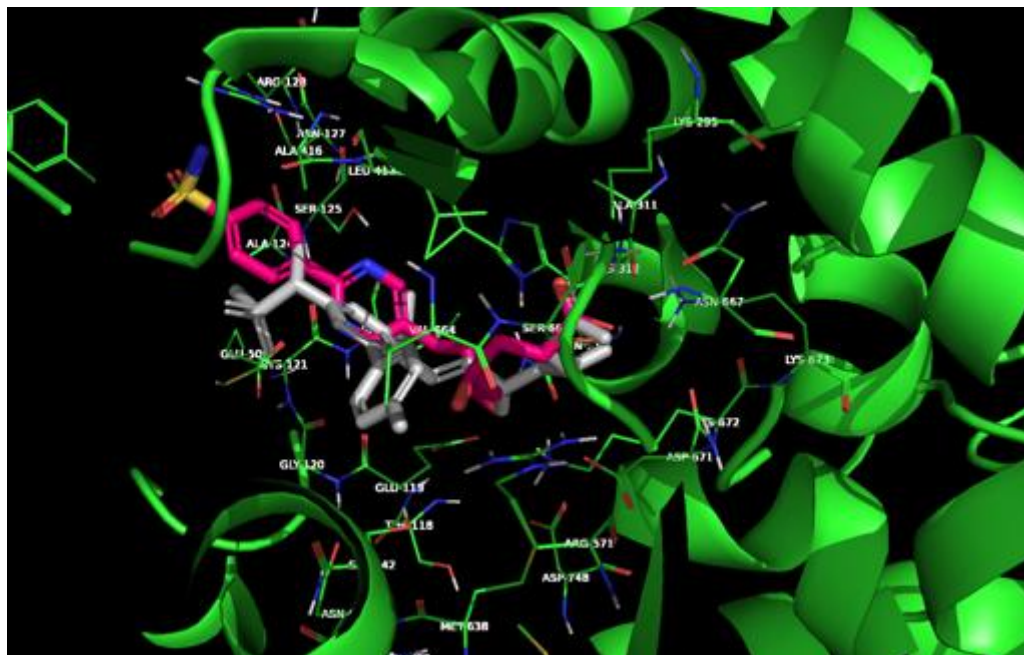


Supporting Figure 37. Best pose for zero-shot with suggested fragments generated molecules with Gemini in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

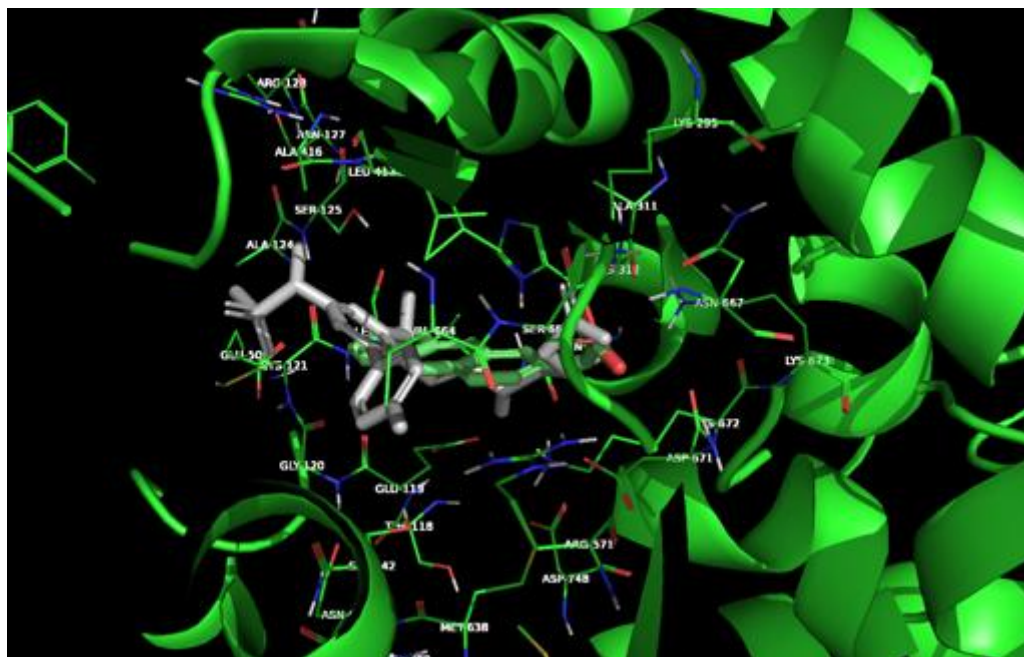


Supporting Figure 38. Best pose for one-shot generated molecules with Gemini in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

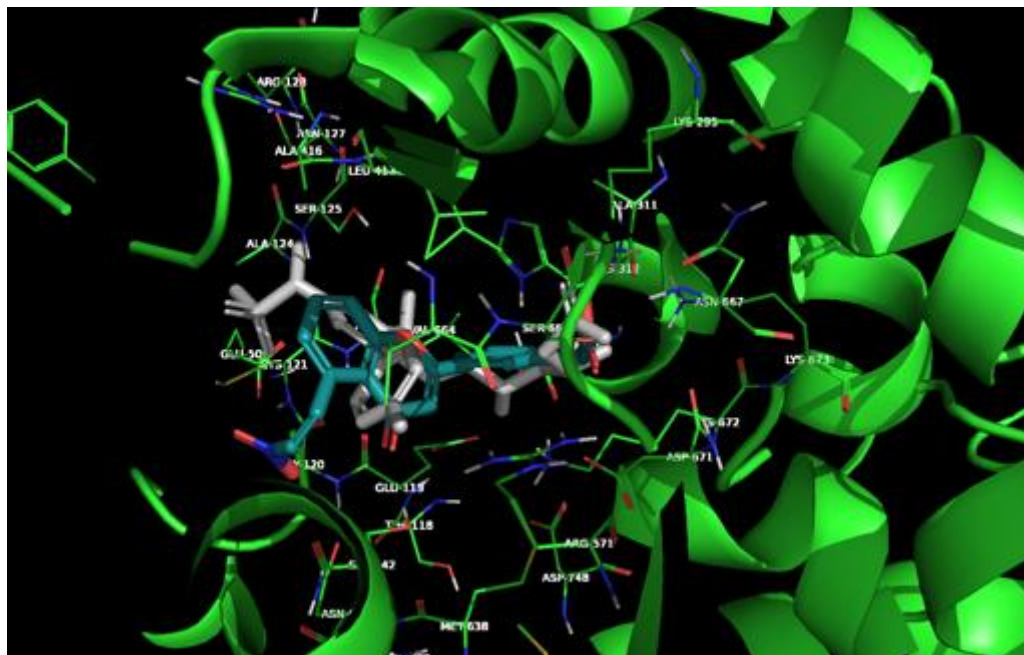
Deepseek-v3.1



Supporting Figure 39. Best pose for zero-shot generated molecules with Deepseek-v3.1 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .



Supporting Figure 40. Best pose for zero-shot with suggested fragments generated molecules with Deepseek-v3.1 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

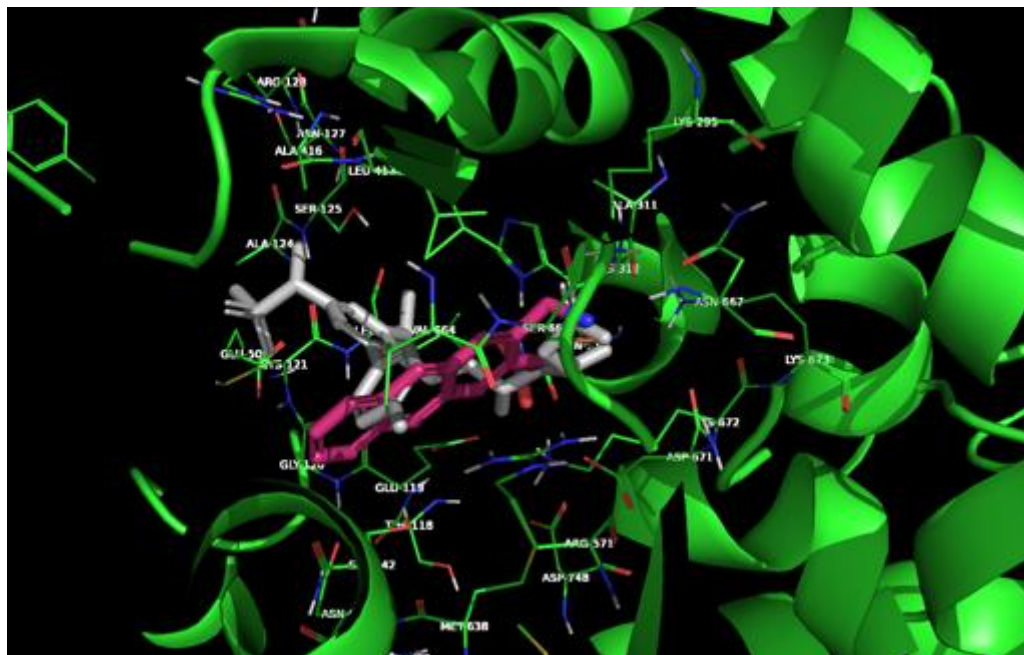


Supporting Figure 41. Best pose for one-shot generated molecules with Deepseek-v3.1 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

Kimi-K2



Supporting Figure 42. Best pose for zero-shot generated molecules with Kimi-K2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

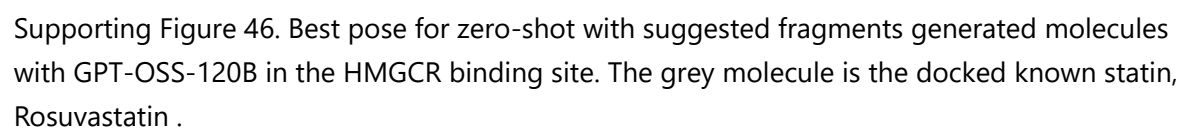
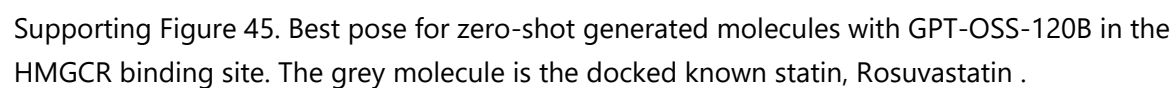


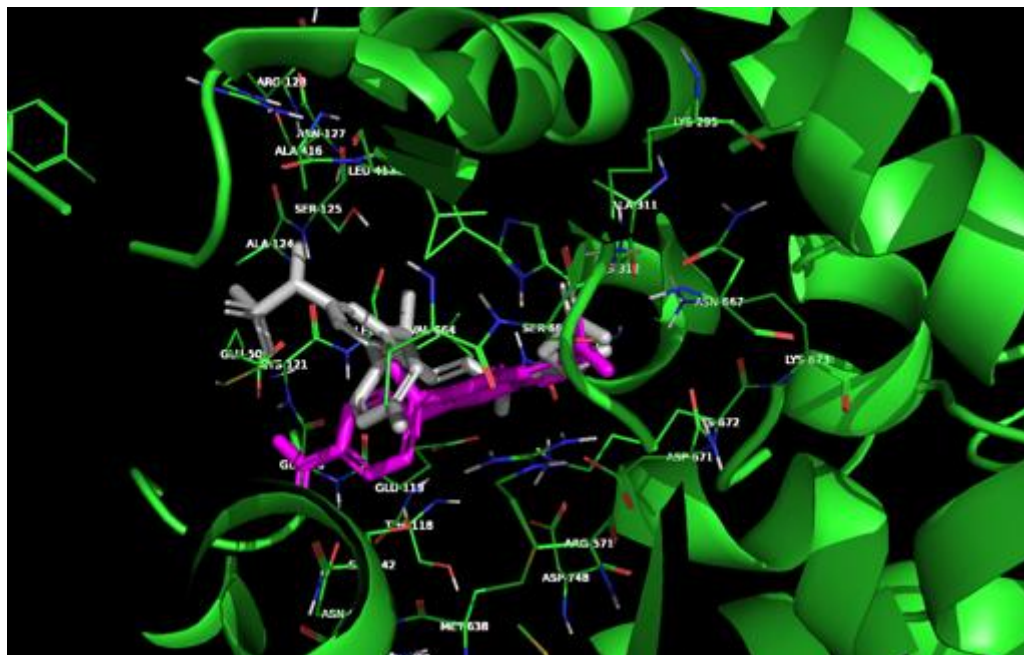
Supporting Figure 43. Best pose for zero-shot with suggested fragments generated molecules with Kimi-K2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .



Supporting Figure 44. Best pose for one-shot generated molecules with Kimi-K2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

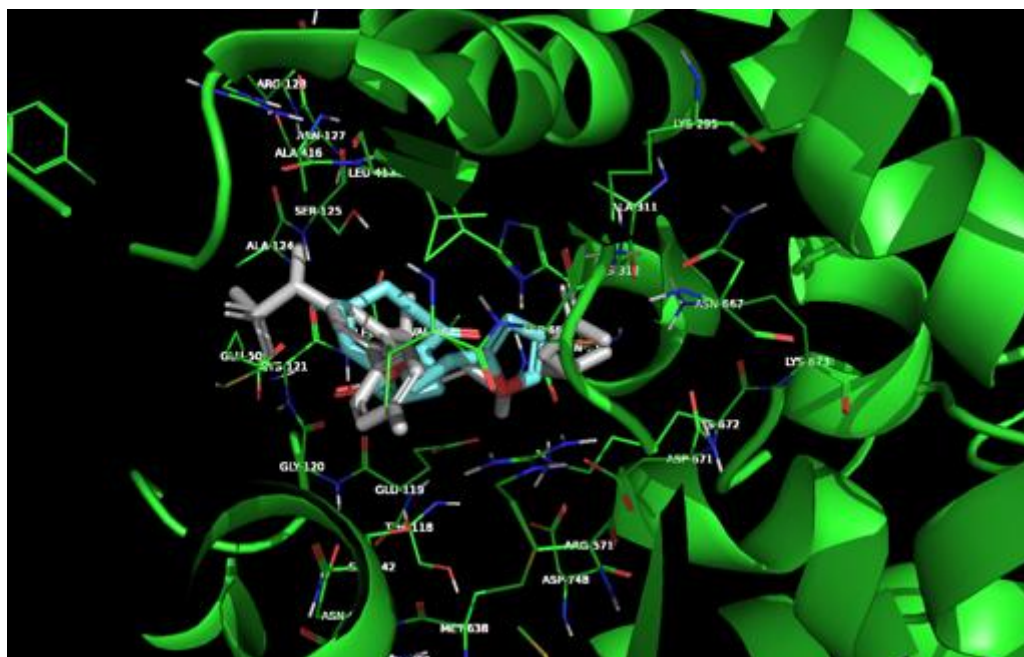
GPT-OSS-120B



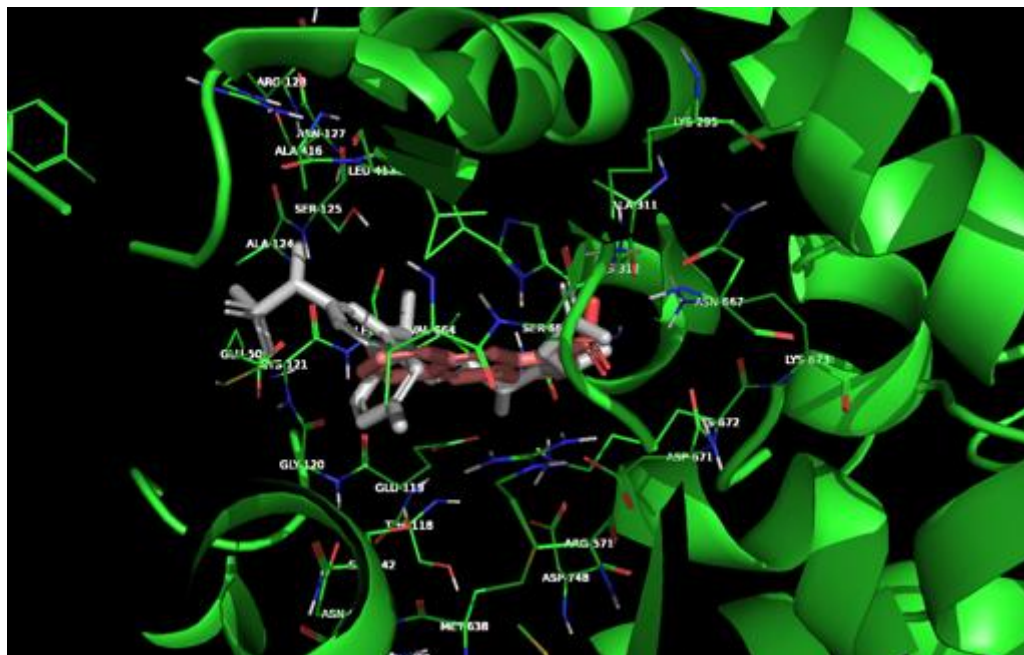


Supporting Figure 47. Best pose for one-shot generated molecules with GPT-OSS-120B in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

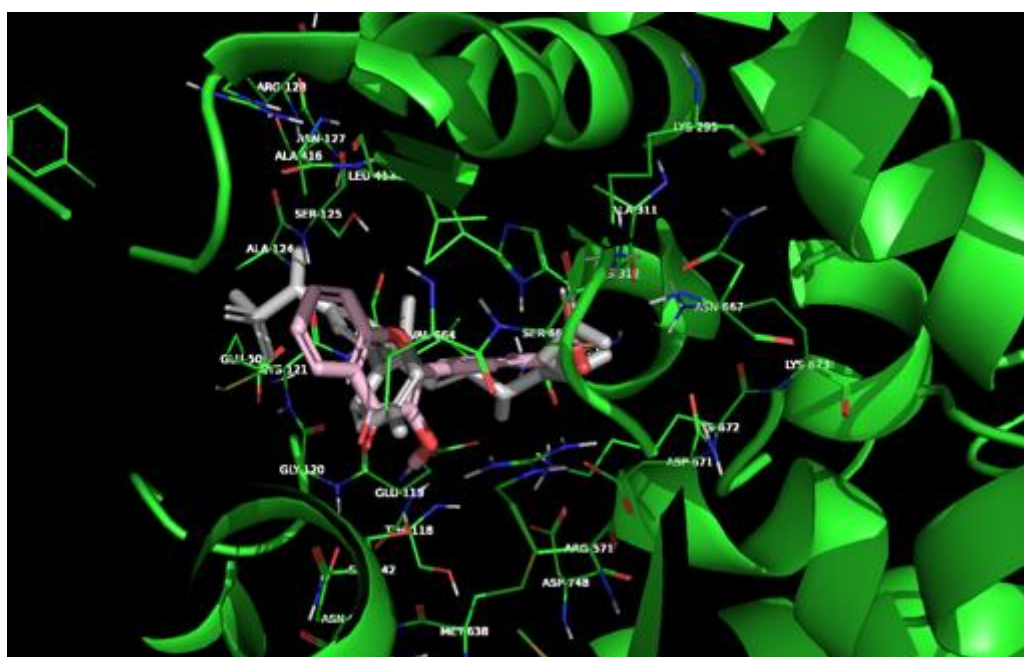
GPT-OSS-20B



Supporting Figure 48. Best pose for zero-shot generated molecules with GPT-OSS-20B in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

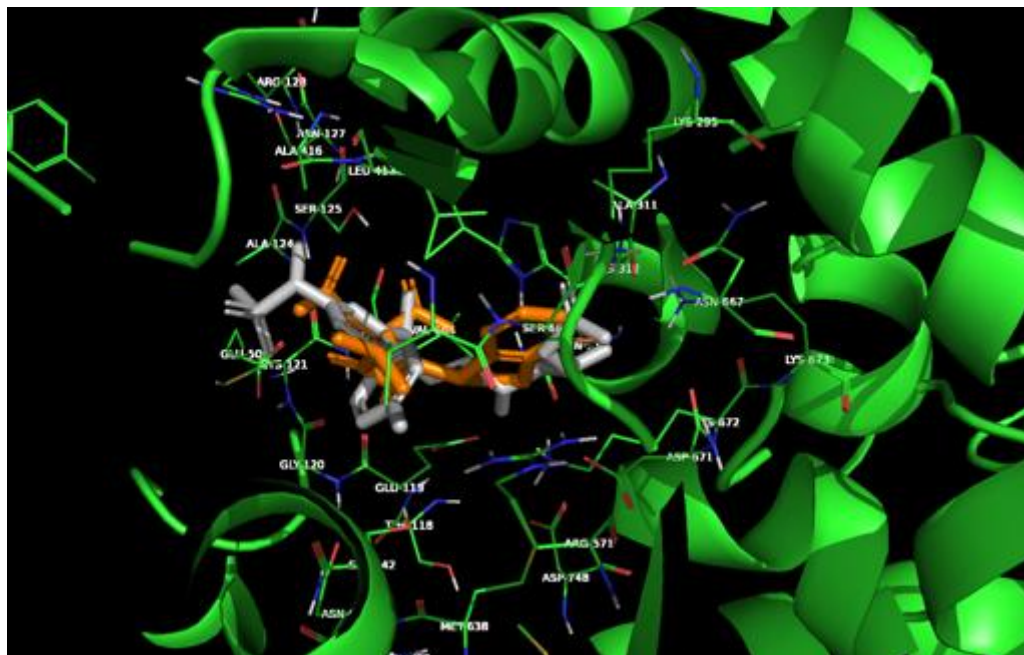


Supporting Figure 49. Best pose for zero-shot with suggested fragments generated molecules with GPT-OSS-20B in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

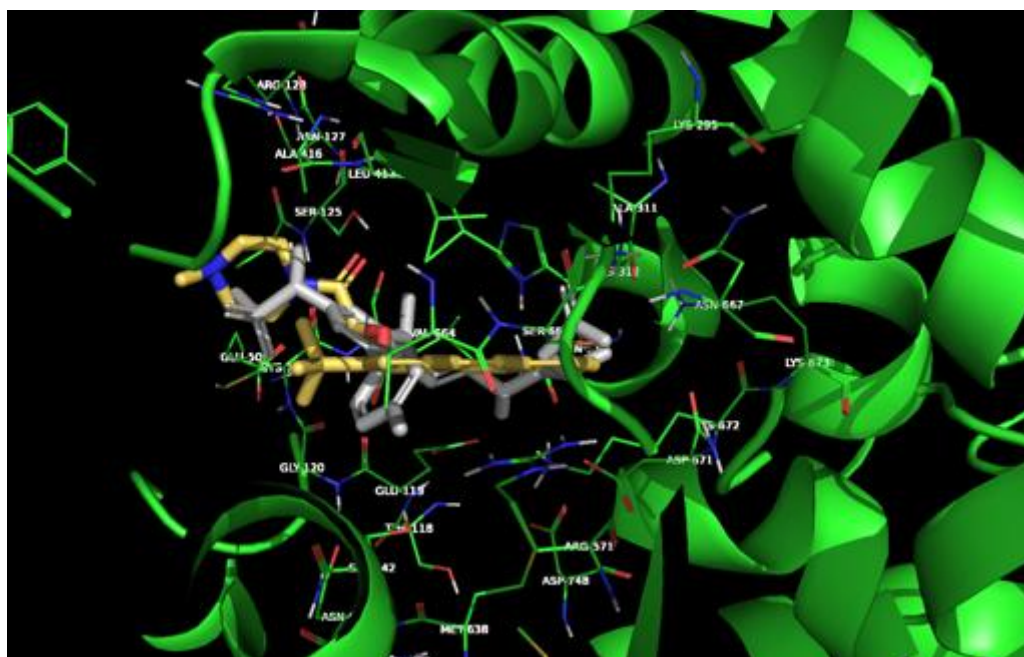


Supporting Figure 50. Best pose for one-shot generated molecules with GPT-OSS-20B in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

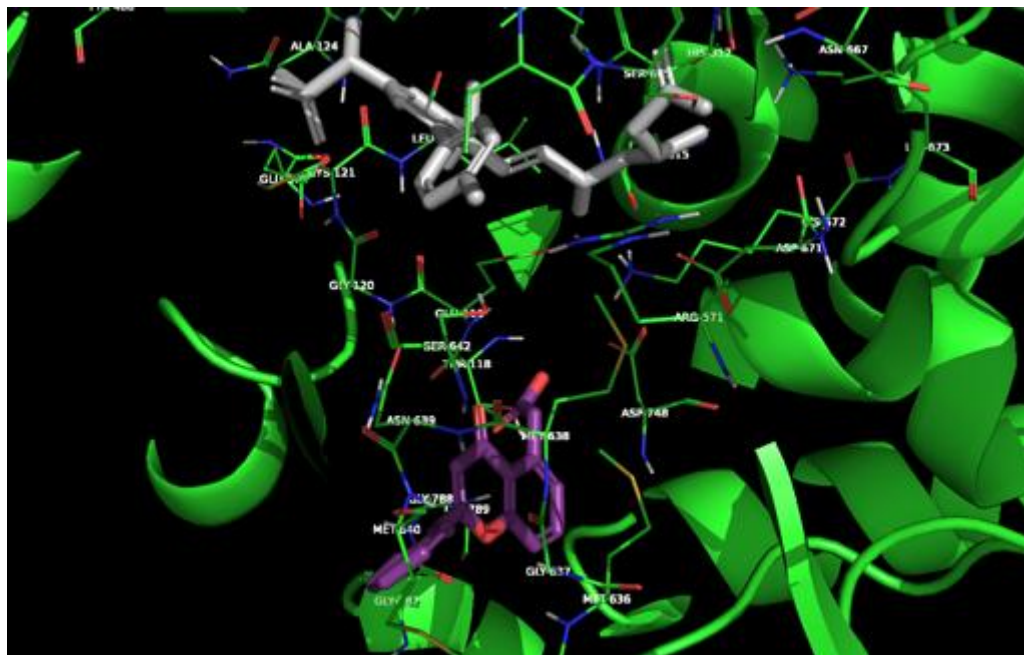
Devstral-2



Supporting Figure 51. Best pose for zero-shot generated molecules with Devstral-2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

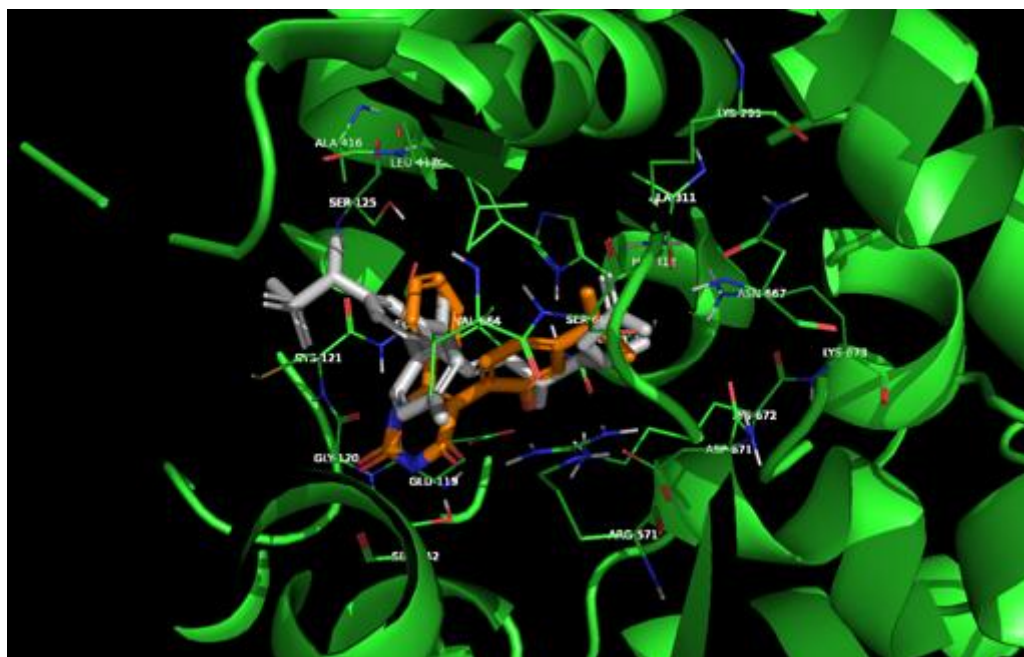


Supporting Figure 52. Best pose for zero-shot with suggested fragments generated molecules with Devstral-2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

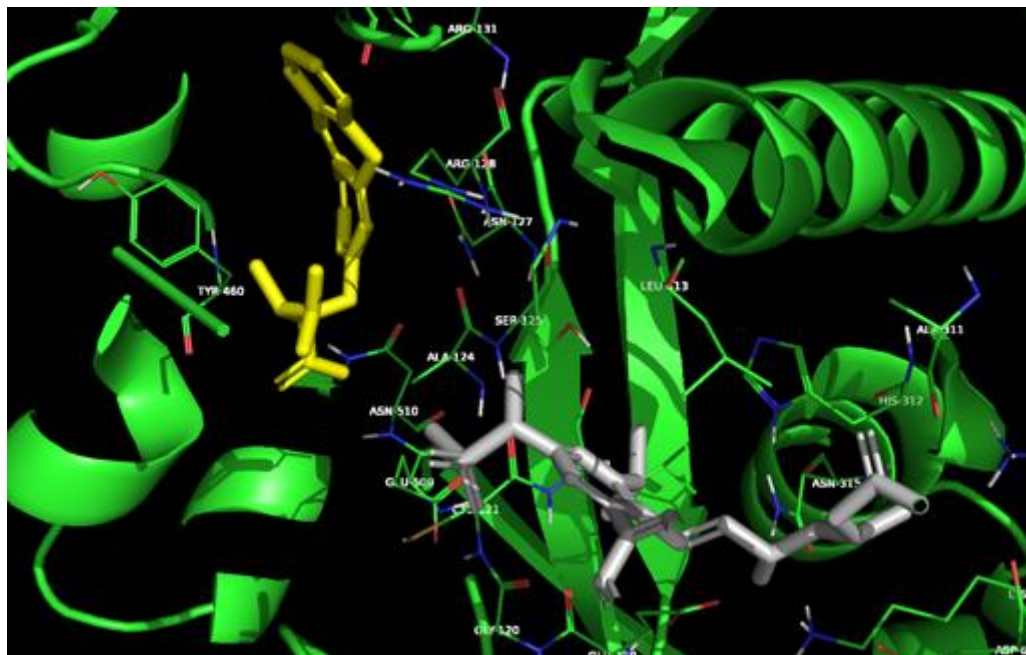


Supporting Figure 53. Best pose for one-shot generated molecules with Devstral-2 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

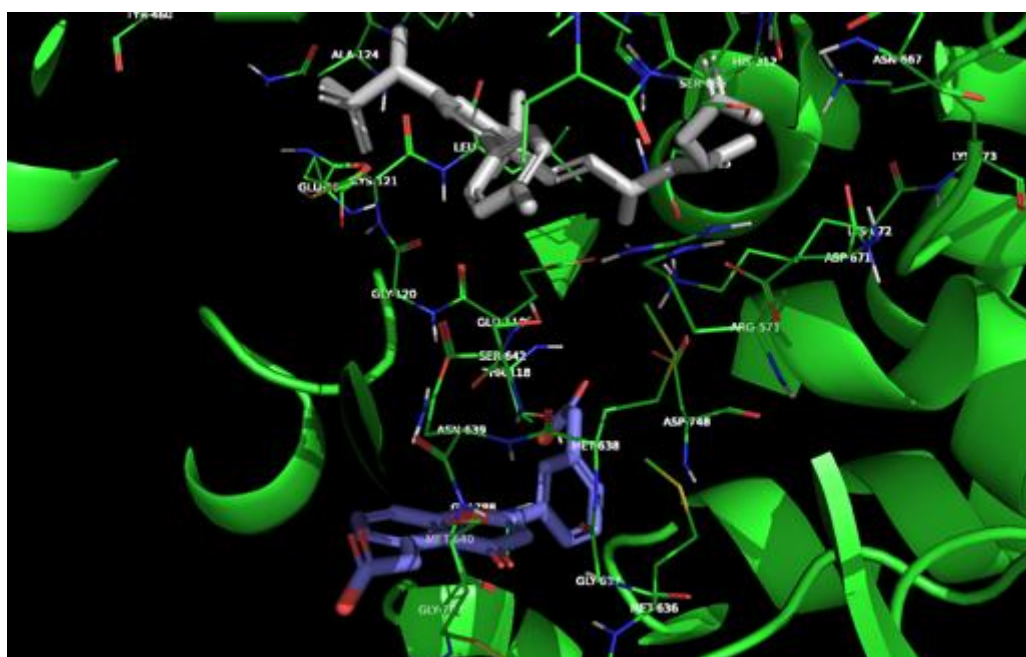
Cogito-2.1



Supporting Figure 54. Best pose for zero-shot generated molecules with Cogito-2.1 in the HMGR binding site. The grey molecule is the docked known statin, Rosuvastatin .

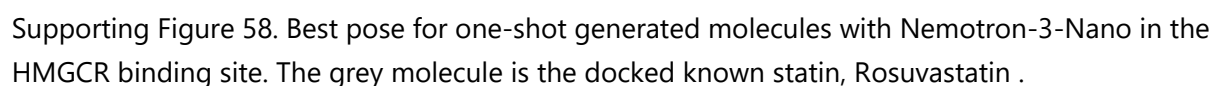
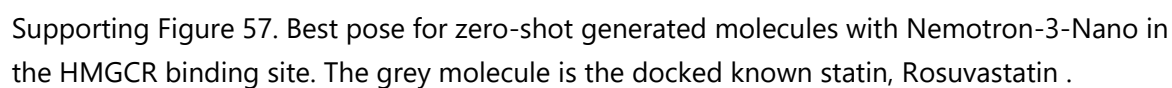


Supporting Figure 55. Best pose for zero-shot with suggested fragments generated molecules with Cogito-2.1 in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .



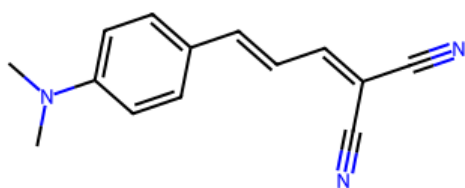
Supporting Figure 56. Best pose for one-shot generated molecules with Cogito-2.1 in the HMGCR binding site. The grey molecule is the docked known statin, Rosuvastatin .

Nemotron-3-Nano

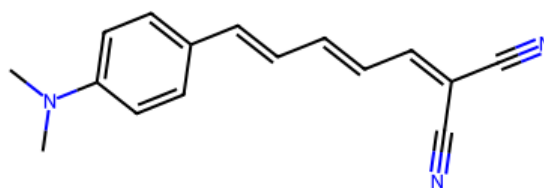


Images of zero-, one-shot and adversarial HOMO-LUMO task molecule generation for all models:

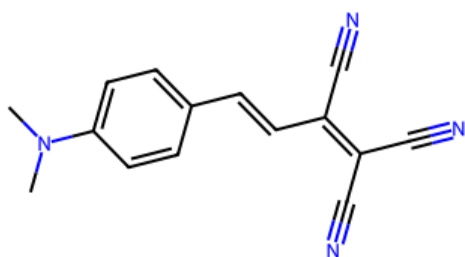
OPENAI



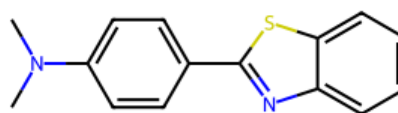
zero_GPT5p2 1



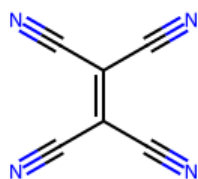
zero_GPT5p2 2



zero_GPT5p2 3

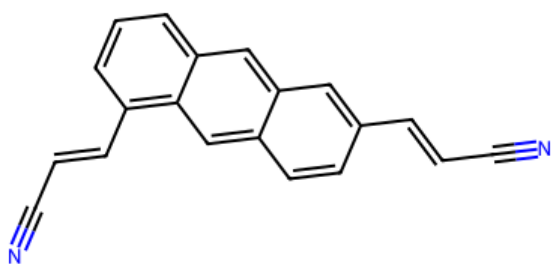


zero_GPT5p2 4

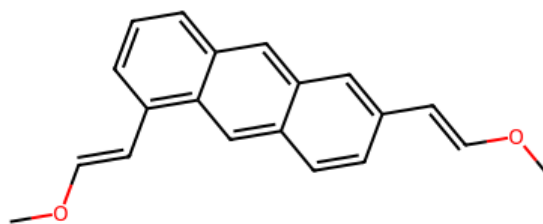


zero_GPT5p2 5

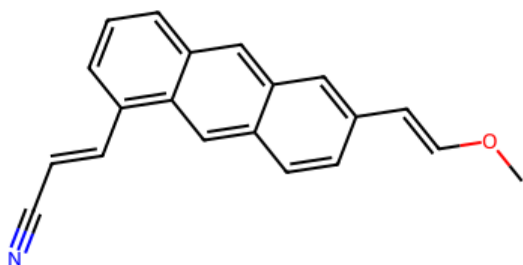
Supporting Figure 59. HOMO-LUMO gap zero-shot generated molecules for OpenAI GPT 5.2.



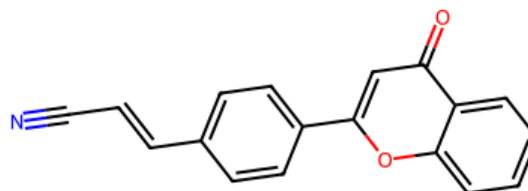
frag_GPT5p2 1



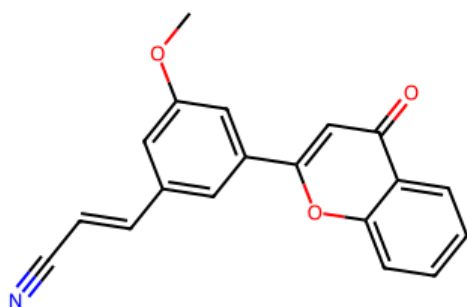
frag_GPT5p2 2



frag_GPT5p2 3

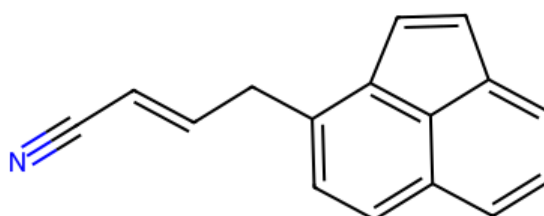


frag_GPT5p2 4



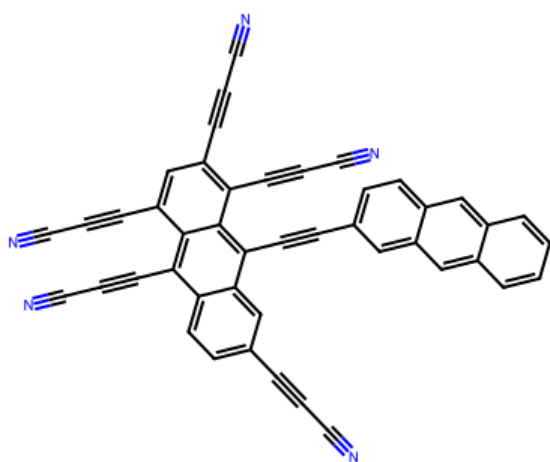
frag_GPT5p2 5

Supporting Figure 60. HOMO-LUMO gap zero-shot with fragments generated molecules for OpenAI GPT 5.2.

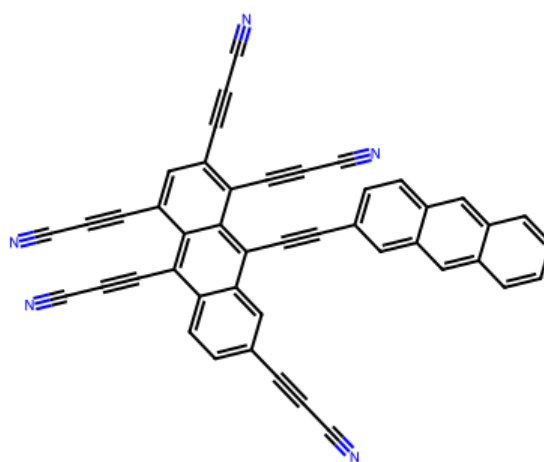


one_GPT5p2 4

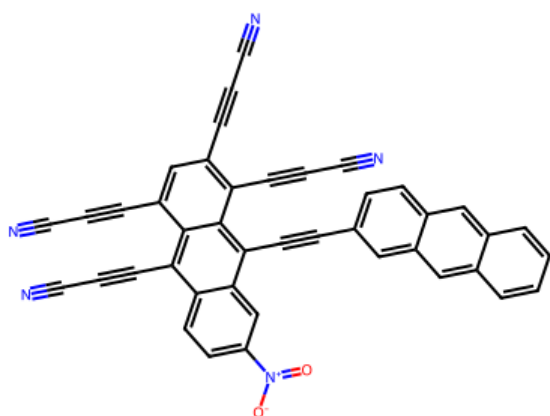
Supporting Figure 61. HOMO-LUMO gap one-shot generated molecules for OpenAI GPT 5.2.



adversarial_GPT5p2 1



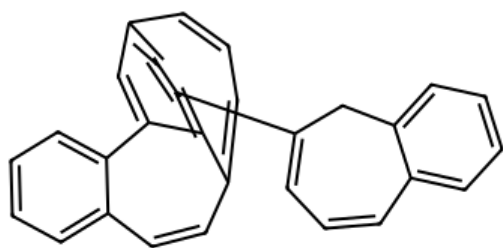
adversarial_GPT5p2 2



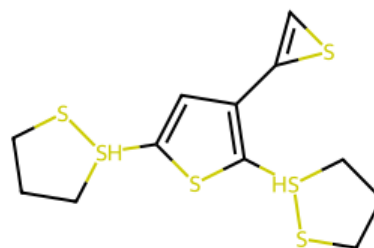
adversarial_GPT5p2 3

Supporting Figure 62. HOMO-LUMO gap adversarial design molecules for OpenAI GPT 5.2.

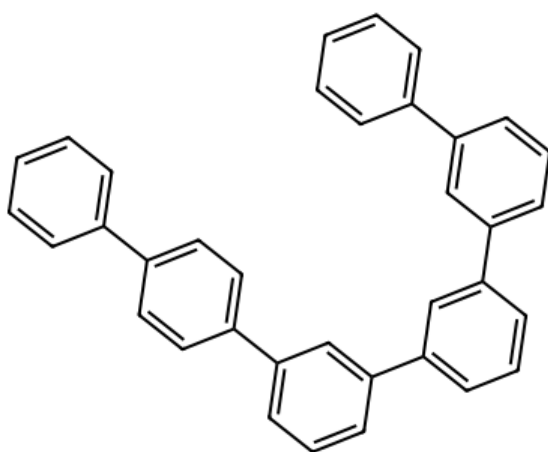
Claude



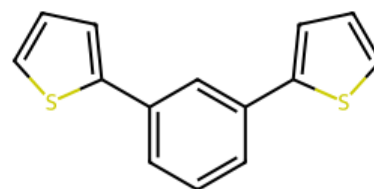
zero_Claude 1



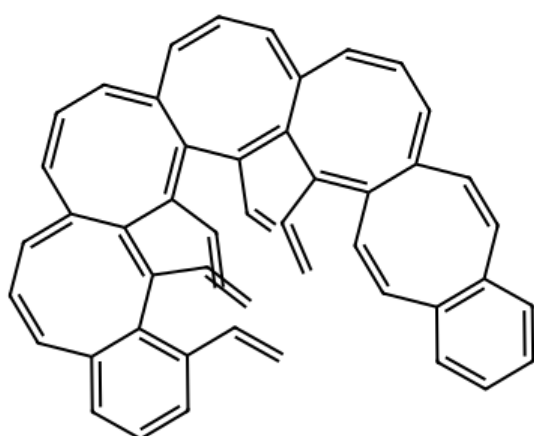
zero_Claude 2



zero_Claude 3

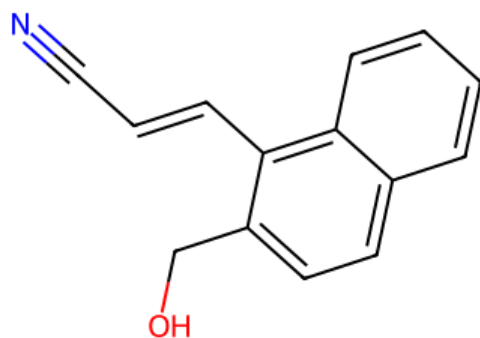


zero_Claude 4

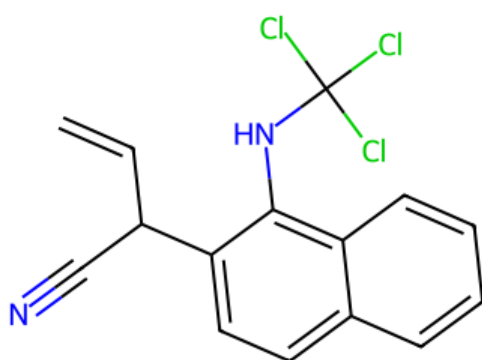


zero_Claude 5

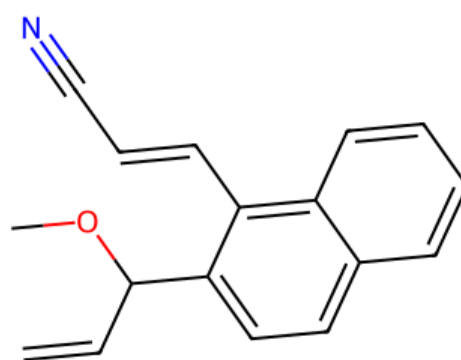
Supporting Figure 63. HOMO-LUMO gap zero-shot generated molecules for Claude.



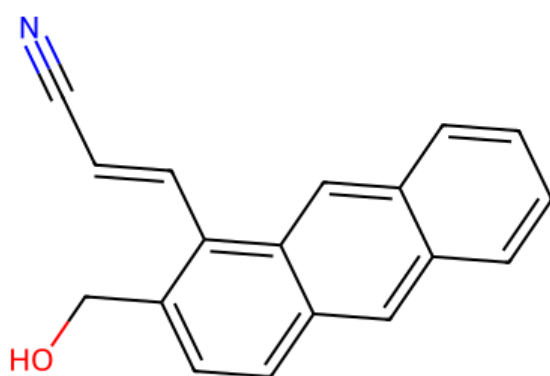
frag_Claude 1



frag_Claude 3

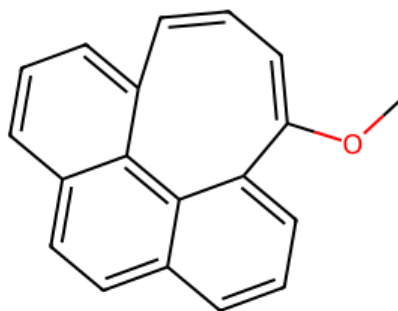


frag_Claude 4

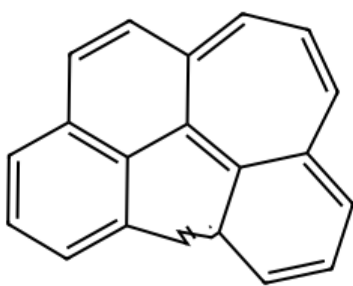


frag_Claude 5

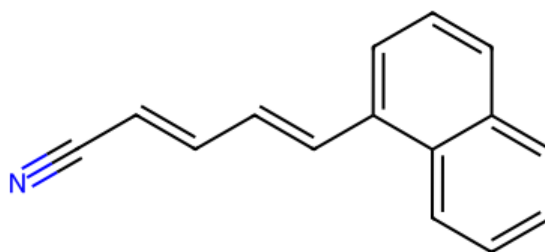
Supporting Figure 64. HOMO-LUMO gap zero-shot with fragments generated molecules for Claude.



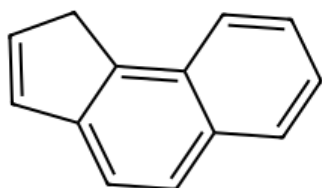
one_Claude 2



one_Claude 3

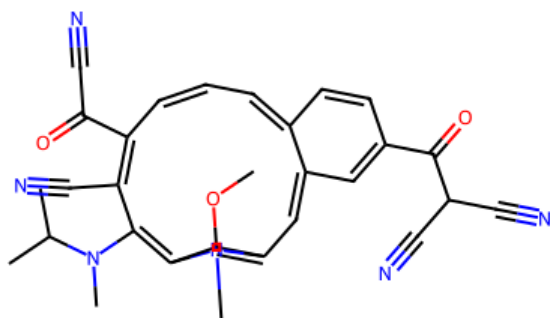


one_Claude 4

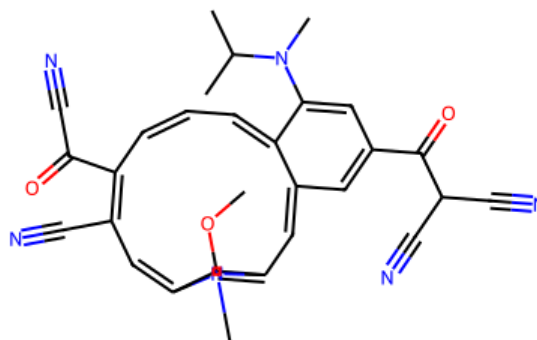


one_Claude 5

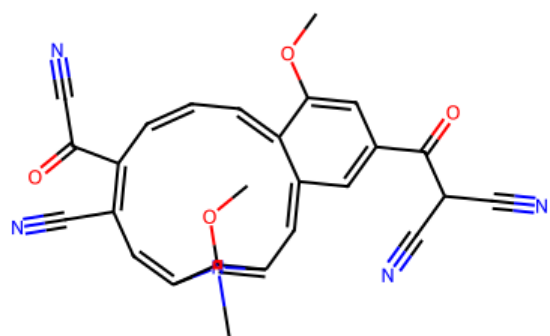
Supporting Figure 65. HOMO-LUMO gap one-shot generated molecules for Claude.



adversarial_Claude 1



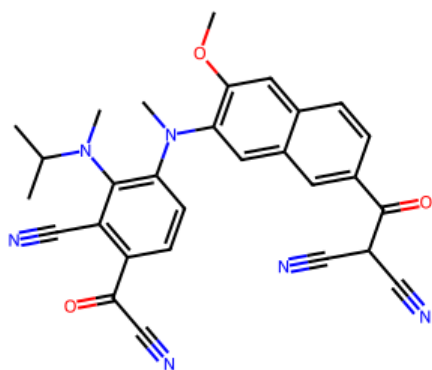
adversarial_Claude 2



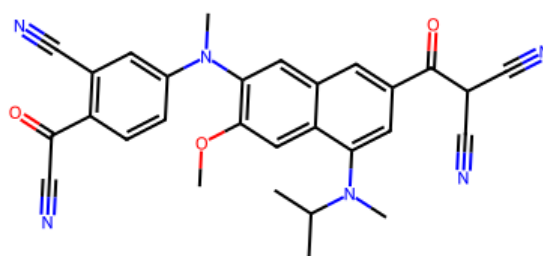
adversarial_Claude 3

Supporting Figure 66. HOMO-LUMO gap adversarial design molecules for Claude.

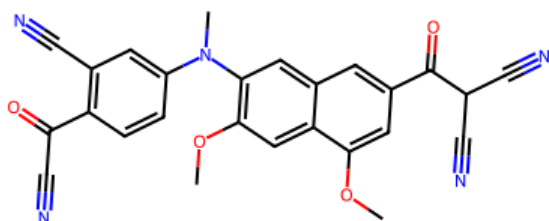
Corrected Claude



adversarial_Corrected_Claude 1



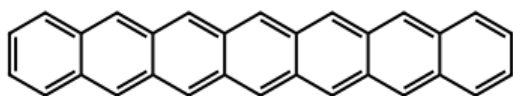
adversarial_Corrected_Claude 2



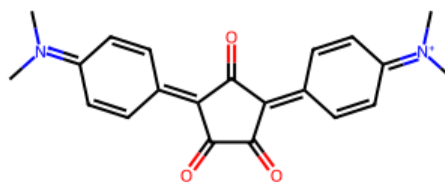
adversarial_Corrected_Claude 3

Supporting Figure 67. HOMO-LUMO gap adversarial design corrected molecules for Claude.

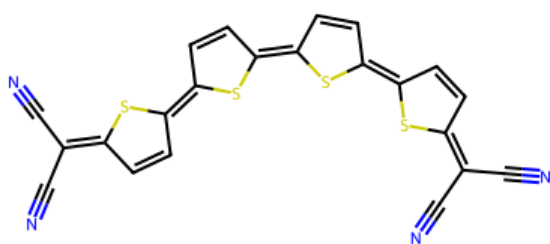
Gemini



zero_Gemini 1

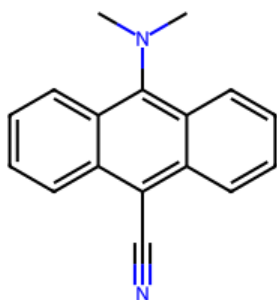


zero_Gemini 4

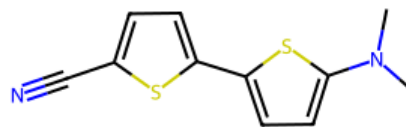


zero_Gemini 5

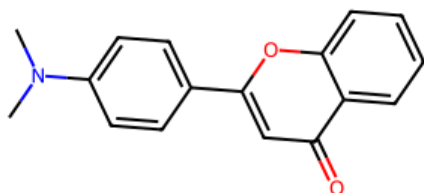
Supporting Figure 68. HOMO-LUMO gap zero-shot generated molecules for Gemini.



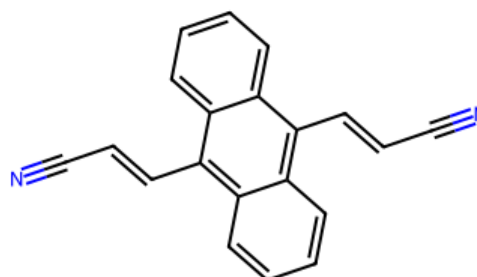
frag_Gemini 1



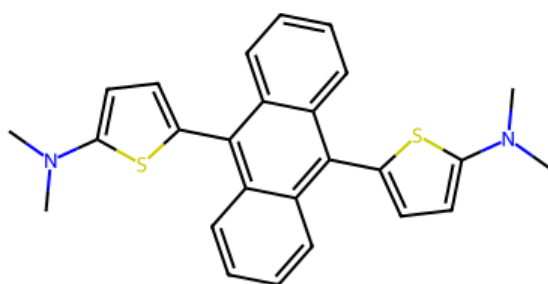
frag_Gemini 2



frag_Gemini 3

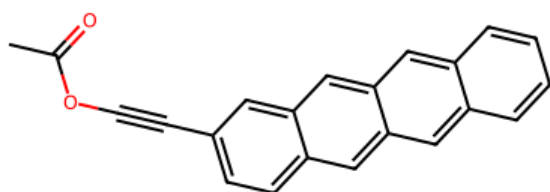


frag_Gemini 4

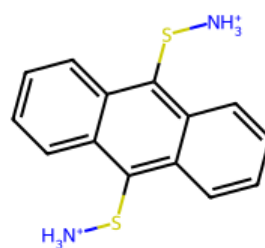


frag_Gemini 5

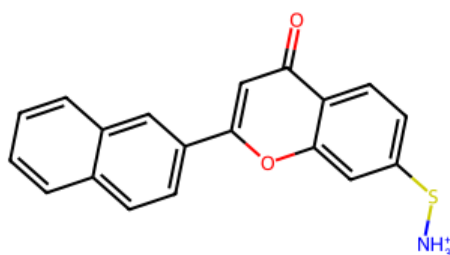
Supporting Figure 69. HOMO-LUMO gap zero-shot with fragments generated molecules for Gemini.



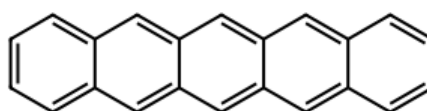
one_Gemini 1



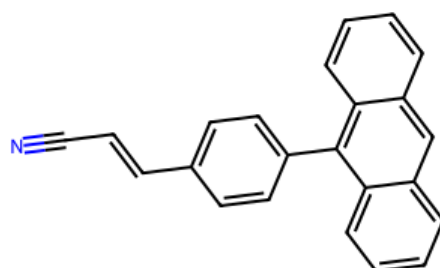
one_Gemini 2



one_Gemini 3



one_Gemini 4

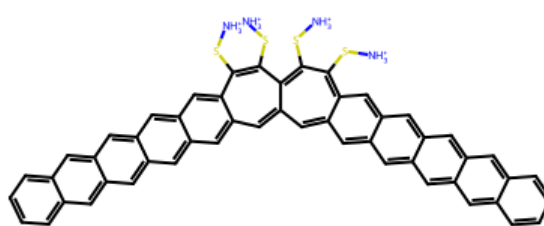


one_Gemini 5

Supporting Figure 70. HOMO-LUMO gap one-shot generated molecules for Gemini.



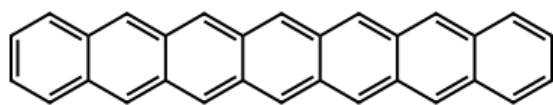
adversarial_Gemini 1



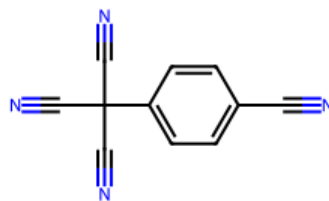
adversarial_Gemini 2

Supporting Figure 71. HOMO-LUMO gap adversarial design molecules for Gemini.

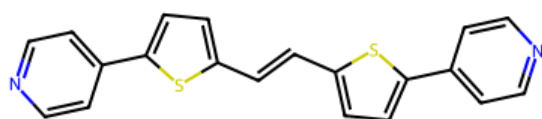
Deepseek



zero_Deepseek 1

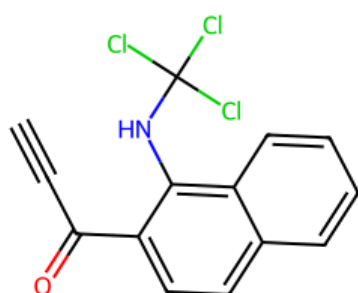


zero_Deepseek 2

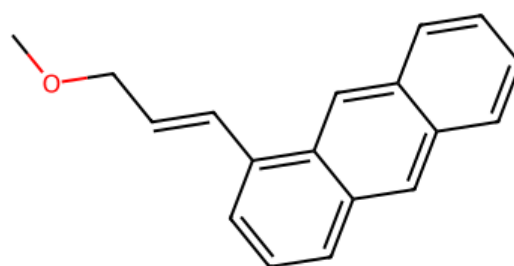


zero_Deepseek 3

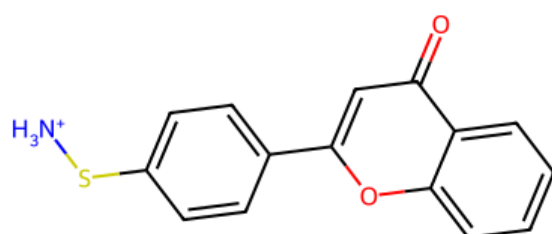
Supporting Figure 72. HOMO-LUMO gap zero-shot generated molecules for Deepseek.



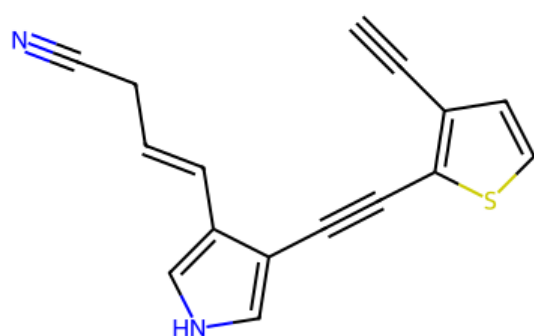
frag_Deepseek 1



frag_Deepseek 2

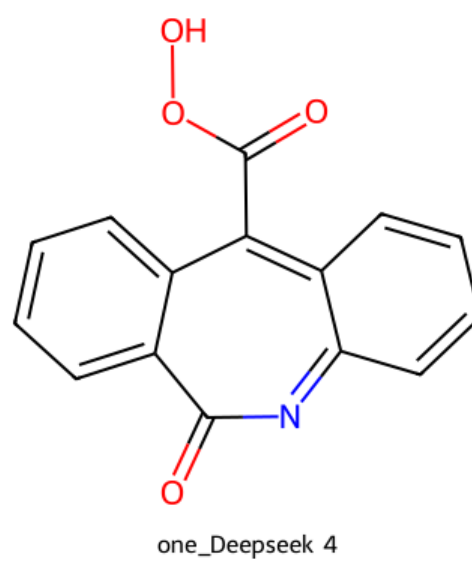


frag_Deepseek 3



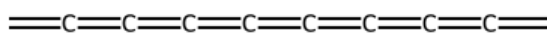
frag_Deepseek 5

Supporting Figure 73. HOMO-LUMO gap zero-shot with fragments generated molecules for Deepseek.



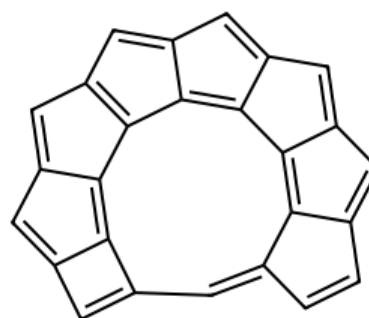
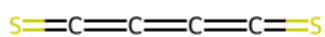
Supporting Figure 74. HOMO-LUMO gap one-shot generated molecules for Deepseek.

Kimi-K2



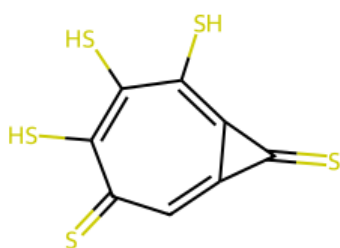
zero_Kimi_k2 1

zero_Kimi_k2 2



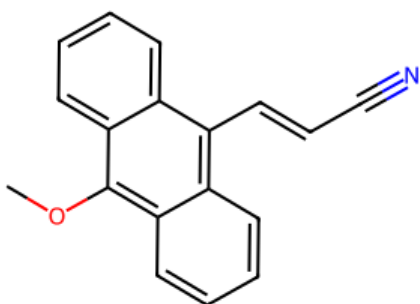
zero_Kimi_k2 3

zero_Kimi_k2 4

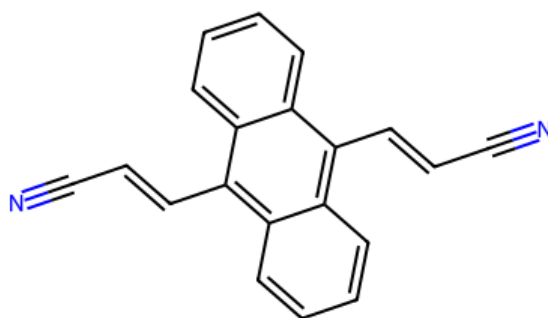


zero_Kimi_k2 5

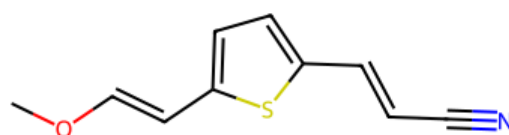
Supporting Figure 75. HOMO-LUMO gap zero-shot generated molecules for Kimi-K2.



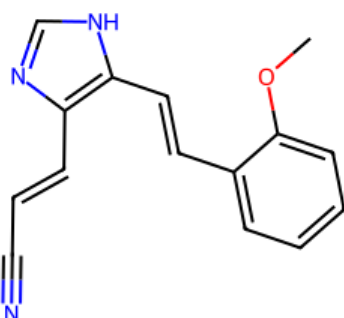
frag_Kimi_k2 1



frag_Kimi_k2 2

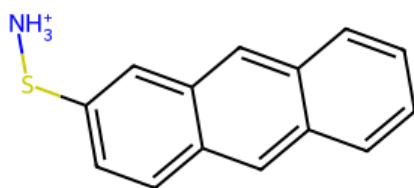


frag_Kimi_k2 4

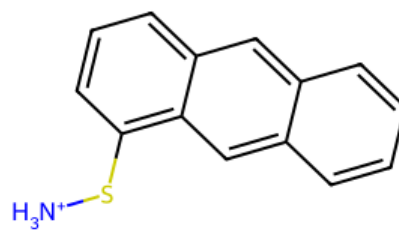


frag_Kimi_k2 5

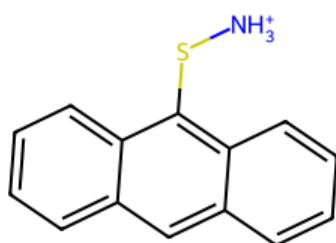
Supporting Figure 76. HOMO-LUMO gap zero-shot with fragments generated molecules for Kimi-K2.



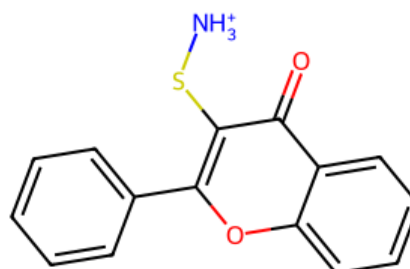
one_Kimi_k2 1



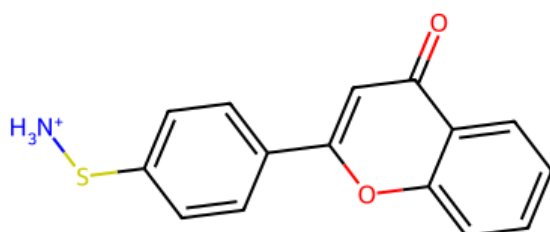
one_Kimi_k2 2



one_Kimi_k2 3



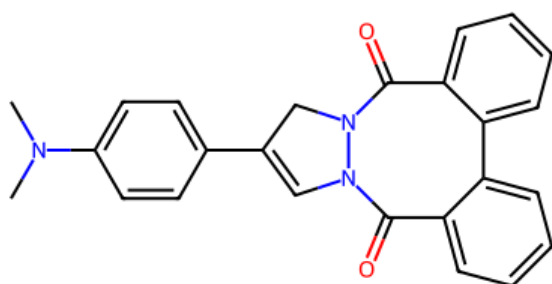
one_Kimi_k2 4



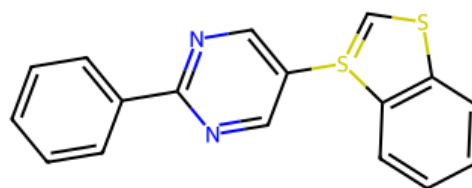
one_Kimi_k2 5

Supporting Figure 77. HOMO-LUMO gap one-shot generated molecules for Kimi-K2.

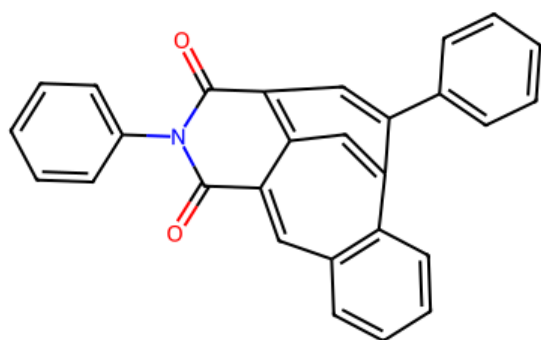
GPT-OSS-120B



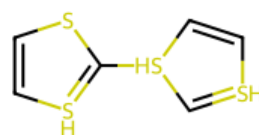
zero_GPT_OSS120 1



zero_GPT_OSS120 2

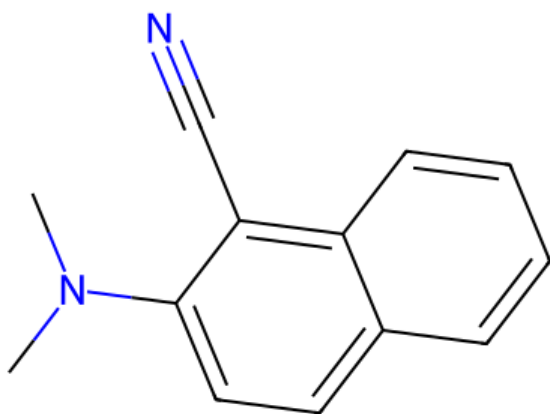


zero_GPT_OSS120 3

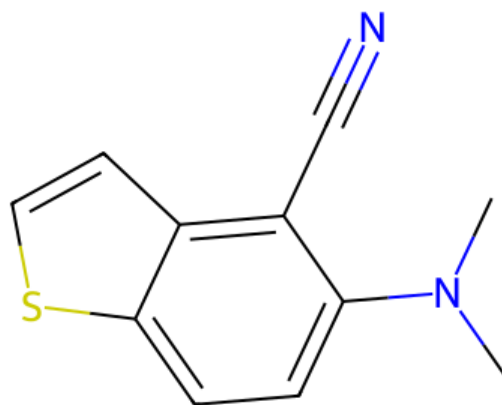


zero_GPT_OSS120 4

Supporting Figure 78. HOMO-LUMO gap zero-shot generated molecules for GPT-OSS-120B.



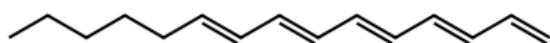
frag_GPT_OSS120 1



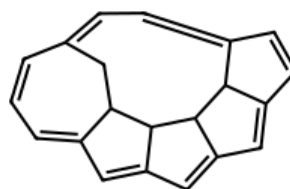
frag_GPT_OSS120 4

Supporting Figure 79. HOMO-LUMO gap zero-shot with fragments generated molecules for GPT-OSS-120B.

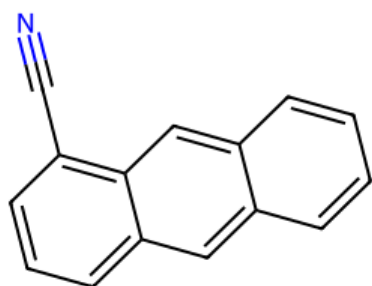
GPT-OSS-20B



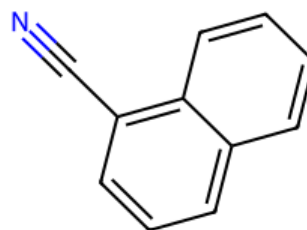
zero_GPT_OSS20 4



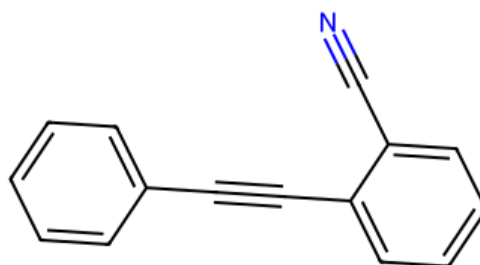
56 / 65



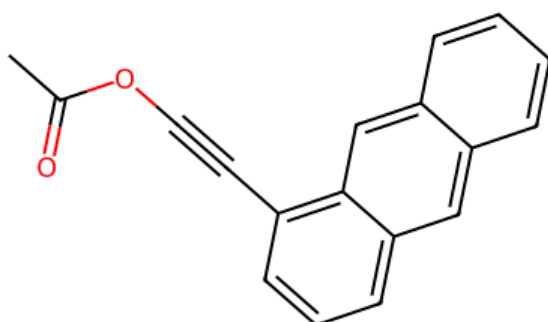
frag_GPT_OSS20 1



frag_GPT_OSS20 2

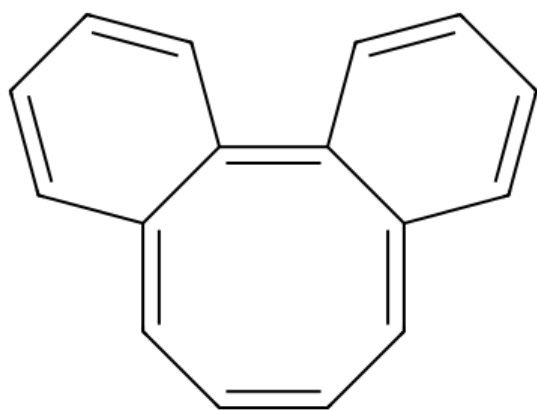


frag_GPT_OSS20 4

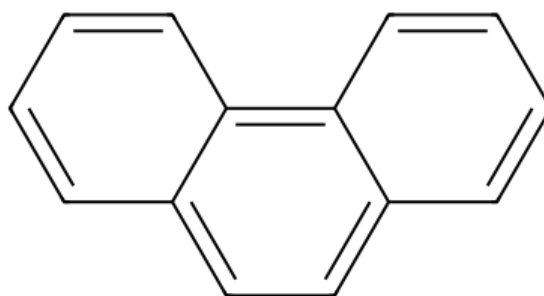


frag_GPT_OSS20 5

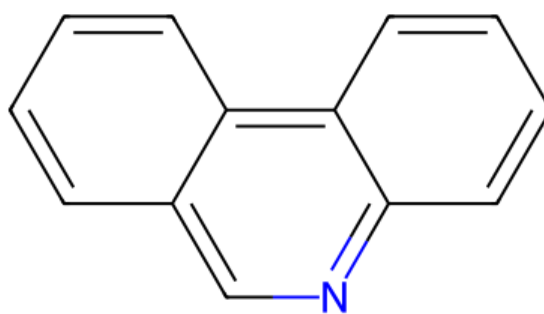
Supporting Figure 81. HOMO-LUMO gap zero-shot with fragments generated molecules for GPT-OSS-20B.



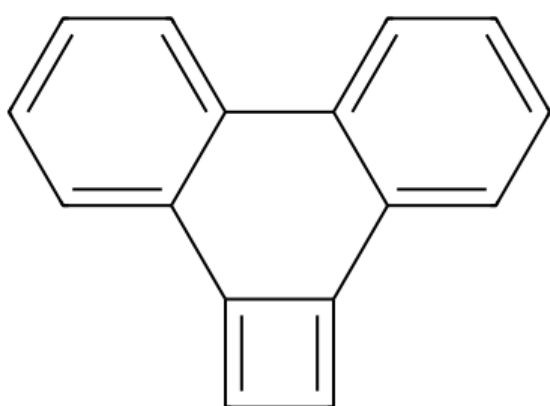
one_GPT_OSS20 1



one_GPT_OSS20 2



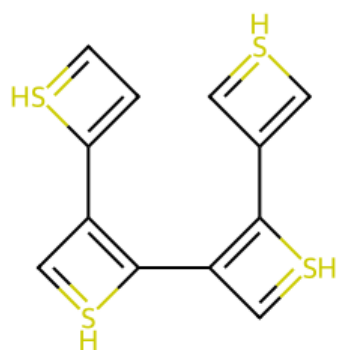
one_GPT_OSS20 4



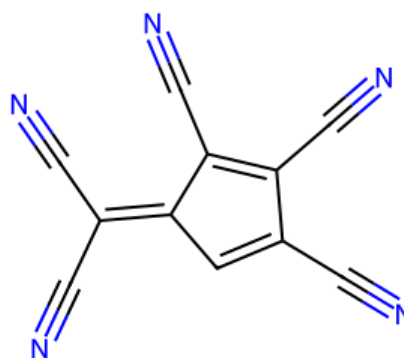
one_GPT_OSS20 5

Supporting Figure 82. HOMO-LUMO gap one-shot generated molecules for GPT-OSS-20B.

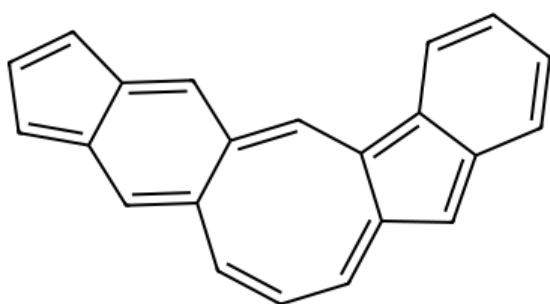
Devstral



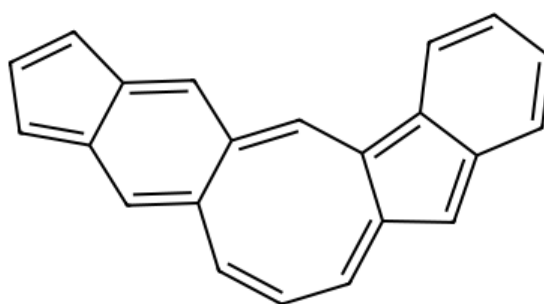
zero_Devstral 1



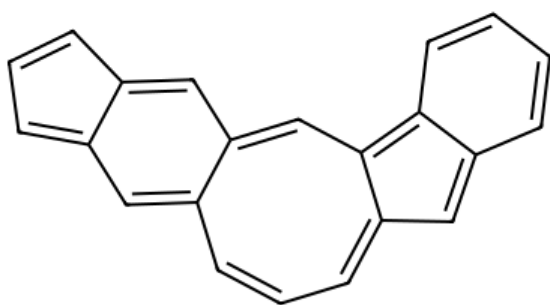
zero_Devstral 2



zero_Devstral 3

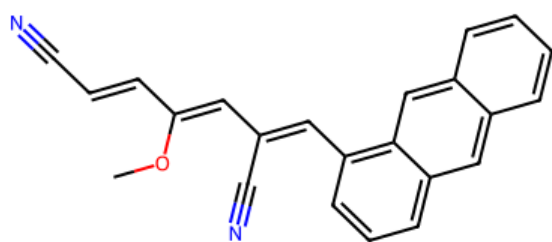


zero_Devstral 4

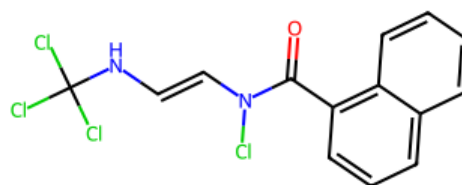


zero_Devstral 5

Supporting Figure 83. HOMO-LUMO gap zero-shot generated molecules for Devstral.

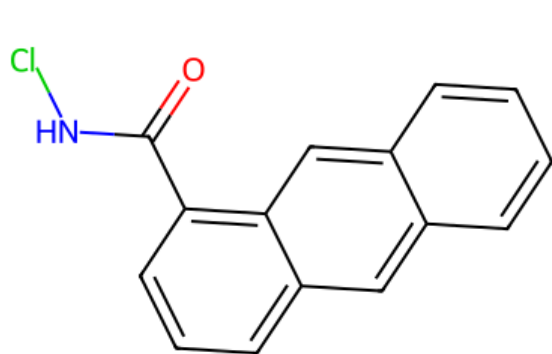


frag_Devstral 1

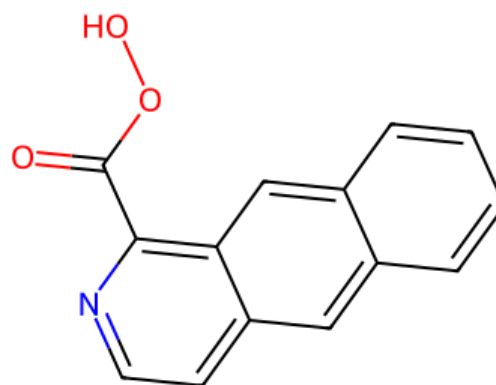


frag_Devstral 4

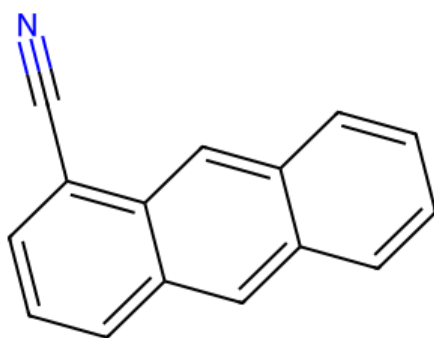
Supporting Figure 84. HOMO-LUMO gap zero-shot with fragments generated molecules for Devstral.



one_Devstral 1



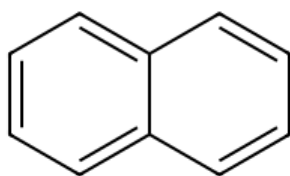
one_Devstral 2



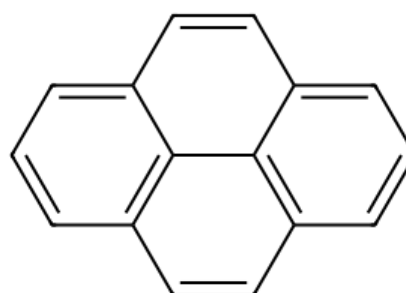
one_Devstral 3

Supporting Figure 85. HOMO-LUMO gap one-shot generated molecules for Devstral.

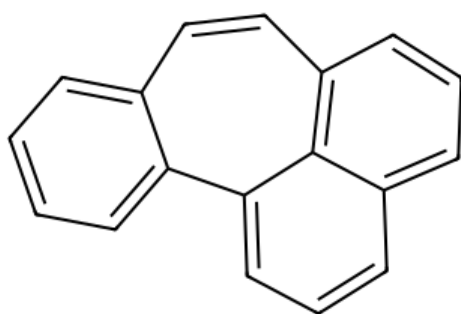
Cogito



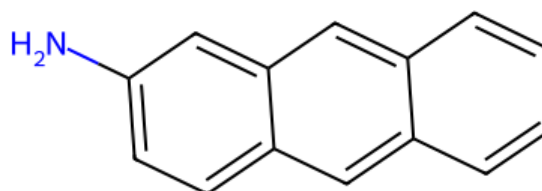
zero_Cogito 1



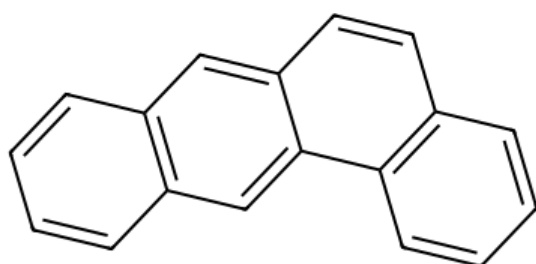
zero_Cogito 2



zero_Cogito 3

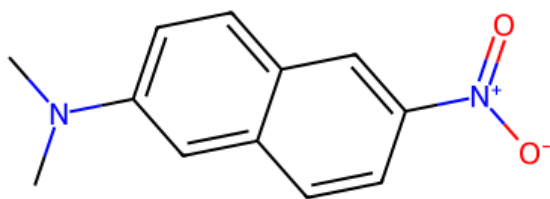


zero_Cogito 4

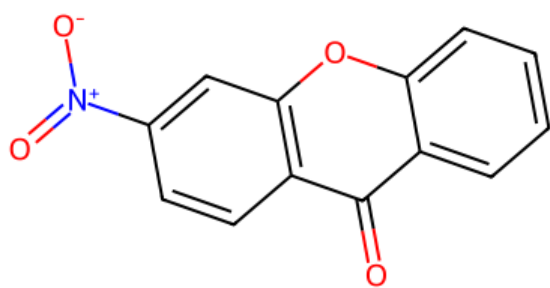


zero_Cogito 5

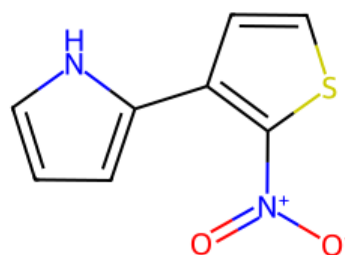
Supporting Figure 86. HOMO-LUMO gap zero-shot generated molecules for Cogito.



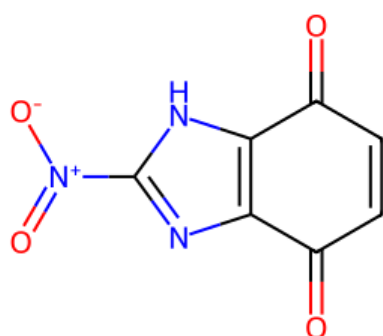
frag_Cogito 1



frag_Cogito 3

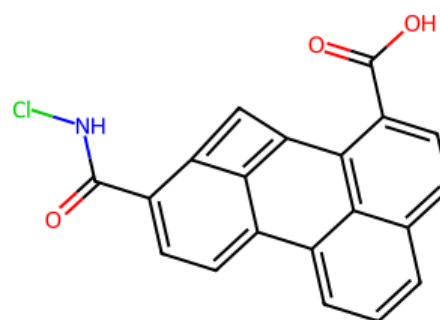


frag_Cogito 4

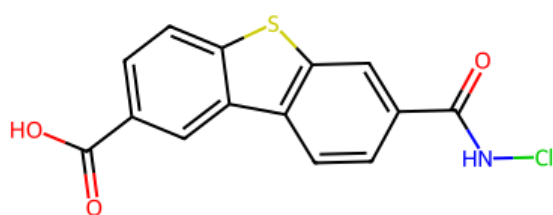


frag_Cogito 5

Supporting Figure 87. HOMO-LUMO gap zero-shot with fragments generated molecules for Cogito.



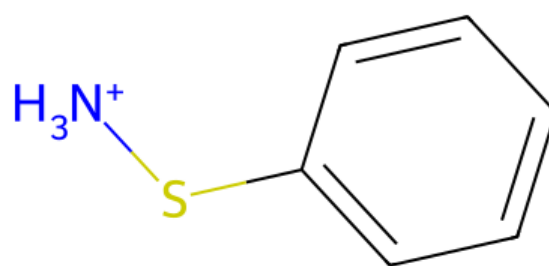
one_Cogito 2



one_Cogito 5

Supporting Figure 88. HOMO-LUMO gap one-shot generated molecules for Cogito.

Nemotron



one_Nemotron 2

Supporting Figure 89. HOMO-LUMO gap one-shot generated molecules for Nemotron.